

Program overview

19-May-2025 14:32

Year 2024/2025
Organization Industrial Design Engineering
Education Master Strategic Product Design

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Strategic Product Design							
1st year Master Strategic Product Design							
Faculty Core Courses							
IDEM4101	Delft Lectures on Design	5					
IDEM4201	Design Processes and Practices	5					
SPD Program Courses							
IDEM3100	Strategic Design Studio	10					
IDEM3101	Foresight and Backcasting	5					
IDEM3102	Research Methods for Strategic Design	5					
IDEM3103	Deep Dive into Organisations	5					
SPD Program Electives (minimum of 15 EC)							
IDEM3201	Qualitative Research Methods for Strategic Design	5					
IDEM3202	Quantitative Research Methods for Strategic Design	5					
IDEM3211	Technology and Strategic Design	5					
IDEM3212	Tech-enabled Entrepreneurship	5					
IDEM3213	Brand Experience Across Technologies	5					
IDEM3221	Societal Missions: Theory and Practice	5					
IDEM3222	Responsible Marketing	5					
IDEM3223	Sustainable Consumer Behaviour	5					
Design Studio Spring Semester (minimum of 10 EC)							
IDEM3210	Tech-enabled Innovation Studio	10					
IDEM4220	Systemic Design Studio	10					
2nd year Master Strategic Product Design							
Electives (minimum of 30 EC)							
IDEM101	Advanced Service Thinking and Design Practice	5					
IDEM102	Build Your Start-up 1: Idea Validation	5					
IDEM103	Build Your Start-up 2: Prototyping	5					
IDEM104	Creative Facilitation	5					
IDEM105	Exploring Design Intelligence	5					
IDEM201	Advanced Visualisation for Communication	5					
IDEM202	Advanced Visualisation for Design	5					
IDEM203	Biomechanics	5					
IDEM204	Child and Play Perspectives	5					
IDEM205	Cognitive Ergonomics for Complex Systems	5					
IDEM206	Contextmapping Skills	5					
IDEM208	Design in Health: Theory & Practice	5					
IDEM210	eHealth Design for a Healthy Society	5					
IDEM212	Food and Eating Design	5					
IDEM214	Lifestyle Research and Design	5					
IDEM215	Lighting Design	5					
IDEM216	PerForm the Unseen	5					
IDEM217	Strategic Automotive	5					
IDEM218	Systemic Automotive Design	5					
IDEM219	Videography	5					
IDEM220	When Images Remain	5					
IDEM221	Making and Prototyping Skills	5					
IDEM301	Animated Materials	5					
IDEM303	Computational Fabrication and MetaMaterials	5					
IDEM304	Design for Emerging Markets	5					
IDEM305	Designing Responsible AI	5					
IDEM307	Generative AI and Design	5					
IDEM308	Graduation Launchpad	5					
IDEM309	Repair & Recycling!	5					
IDEM401	BlueDot: Design Contest	3					
IDEM402	BlueDot: Organisation	5					
IDEM403	Research (3 ECTS)	3					
IDEM404	Research (5 ECTS)	5					
SPD Graduation Project (30 EC)							
ID4390-16	Graduation Project (SPD)	30					
IDEM3000	Graduation Project SPD	30					

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Master Strategic Product Design

Introduction 4

A mandatory application for all courses (mandatory and electives courses) shall be made via the electronic applications system Osiris in the period that Osiris there to is opened. This period closes about 5 calendar weeks prior to the first day of the semester in which the programme starts; the faculty announces the exact deadline for application timely. The application for courses has to take place per semester, meaning for courses of 2 quarters at once.

See <http://www.io.tudelft.nl/osiris> for deadlines and details.

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1st year Master Strategic Product Design

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Faculty Core Courses

IDEM4101	Delft Lectures on Design	5
Course Coordinator	V. Tassinari	
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This is a faculty core course for all IDE MSc students TU Delft is a world leading institution in design research. In this course, all first year Design Master students at TU Delft are given the chance to get in touch with the plethora of diverse and cutting edge design research present in the faculty, and further explore its potential for their own future career development. Delft Lectures on Design highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on design practice. The course comprises a series of interactive talks with researchers of the Delft Faculty of Industrial Design Engineering (and/or guest lecturers), who will be addressing how their research insights and expertise link to key contemporary positions in design discourse.	
Study Goals	Upon finishing this course, students are able to: 1. Analyze design research and specific topics in design research. 2. Formulate critical questions justified in the design research literature and other sources. 3. Demonstrate an understanding of the value of particular design research topics to society and the broader field.	
Education Method	The course consists of three cycles of three weeks each, during which students are first introduced by an IDE-researcher to a research theme and topics, followed by individual reflections as well as discussions in small groups. Based on these, each group formulates questions/provocations regarding a specific topic, which are then worked out to be communicated to the researcher in the form of a so-called provotype. A selection of these provotypes will be presented during a plenary event, followed by a moderated, interactive discussion between researcher(s) and students. At the end of the cycle each student individually reflects on the value of the provotype, the present and future relevance of the research as well as on personal lessons learned from the perspective of a future designer/researcher.	
Assessment	The study goals of this course will be assessed by: (pass/fail) Each cycle ends with an individual deliverable in which work of the three weeks, thus including results of the group work, is combined; deliverables C1, C2 and C3. C1 is assessed formative, where C2 and C3 are assessed summative with a pass/fail grading. Both C2 and C3 need to get a pass, to successfully complete the course.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: dld-ide@tudelft.nl	

IDEM4201	Design Processes and Practices	5
Course Coordinator	Dr. P.A. Lloyd	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	This is a faculty core course for all IDE MSc students. Design Processes and Practices introduces you to contemporary theory, methods, principles, and practices of designing in a broad and general sense. Exposing you to many diverse ideas about the design process - from theorists, scientists, methodologists, and practitioners - will allow you to appreciate the full range of factors that are in play when designing. In doing this, the course aims to help you develop a more nuanced vocabulary for talking about and doing the different activities that make up designing. Such a vocabulary will not only help you to understand the practice of design, but also help you to design your final Masters project.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the basic acts of design and which concepts, methods and approaches are appropriate. 2. Articulate theoretical aspects of design processes and practices. 3. Create a design process guide to support your future self in design projects.	
Education Method	The course features lectures, optional seminars, and a broad range of study materials (e.g. books, podcasts, videos, articles, blog posts, academic papers). As you progress through the course, you will individually produce a design process guide suitable to support you through your final graduation project. You will receive formative feedback at several points during the course. During the course, the different phases of the design process are referred to as acts, implying a narrative, dialogical and human structure taking place over a period of time and completed by different actors. Such a structure contains elements of prescription and description, as well as the latest developments regarding non-human actors. Course Structure: Week 1: Introduction and design guide sprint Week 2 & 3: Act one of design process Week 4 & 5: Act two of the design process Week 6 & 7: Act three of the design process Week 8 & 9: Act four of the design process Week 10: Course summary and conclusion	
Assessment	The study goals of this course will be assessed by: - Design Process Guide - Individual exam The final grade is pass/fail constituted by two summative individual assessments. Both assessments must be passed to pass the course.	
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Special Information	Contact: dpp-ide@tudelft.nl	

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SPD Program Courses

IDEM3100	Strategic Design Studio	10
Course Coordinator	R.A. Price	
Contact Hours / Week x/x/x/x	8	
Education Period	1 2	
Start Education	1	
Course Contents	This is a SPD programme core design studio. In this course you will go through all the steps and challenges of a strategic design project, and practice key design capabilities essential for the development of your professional design skills. The course offers a holistic approach to strategic design practice, to prepare you with the knowledge and practical skills necessary for real-world strategic design challenges. Essential elements of the course include understanding the internal and external context of an organisation, while integrating knowledge from multiple perspectives (strategy, organisation and innovation ecosystems, while balancing leadership and ethical considerations), to (re-)frame organisational challenges. You will use collaborative and creative approaches to engage stakeholders, to explore solutions and interventions based on coherent design reasoning, using design leadership to scale the strategic impact of design. Visualisation and making skills will play a prominent role in this course, as you will learn and practice how to use expressive means (e.g., visualisation, prototyping or video making) to communicate your strategic skills in an engaging and collaborative dialogue with relevant stakeholders, and to give tangibility to your outcomes. You will work on real assignments from different kinds of organisations (private, public, non-profit) and/or innovation ecosystems and focus on different kinds of challenges (commercial, societal, technology-related). The briefs you will receive range in different areas of strategic design going from identifying new market opportunities, developing new innovation approaches, investigating how the company can leverage their competences to proposing new visioning projects and disrupting existing organisational norms.	
Study Goals	Upon finishing this course, students are able to: 1. Identify and apply relevant methods and practices to carry out strategic design projects 2. Analyse products/services, organisations and ecosystems with a strategic lens 3. Integrate knowledge on strategy, organisations and ecosystems, as well as leadership and ethical considerations, in order to design viable and feasible pathways for organisations 4. Demonstrate professional skills to manage and deliver projects with external stakeholders 5. Reframe innovation challenges into strategic design opportunities, based on a well-justified design reasoning 6. Reflect in- and on-action during problem- solution exploration 7. Apply visual and tangible expressive skills to convincingly and clearly communicate strategic outcomes to a diverse audience	
Education Method	The course combines (guest) lectures, studio coaching, workshops and field visits. There are three main modules in the course, to allow you a steady development, building up towards more challenging and practice-based design problems. Module 1: [Week 1.2 to Week 1.4 -- 20%]. Case study, lectures, readings and portfolio reflection. In this module, you will reflect on what strategic design is and on your professional identity. Module 2: [Week 1.5 to Week 1.9 -- 30%] Challenge-based learning in a hypothetical context, supported by studio coaching and lectures. Module 2 includes creativity-related workshops (in week 1.5) and media intensive workshops (in weeks 1.6 to 1.8). Module 3 [Week 2.1 to Week 2.10 -- 50%] Client Project: Challenge-based learning with design brief provided by the client, supported by studio coaching and lectures. By having two design projects, you will have the opportunity to build on previous learnings to inform the next.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4350-16 Design Strategy Project.	
Assessment	The study goals of this course will be assessed by: - Assessment module 1 (pass/fail) - Assessment module 2 (pass/fail) - Assessment module 3 (pass/fail) There will be three summative assessment moments (resulting in Pass, Fail or Fail with opportunity to compensate), at the end of each module. These summative assessment methods are supported by multiple formative assessment moments during the studio coaching. Moreover, Module 3 will also include two additional formative assessment methods: a midterm presentation for clients and coaches, and a student peer feedback moment. The distribution between group and individual work across the three summative assessments is the following: - Assessment module 1: group - Assessment module 2: group - Assessment module 3: individual and group There is a stronger emphasis on group work, to reflect the nature of practice and definition of strategic design (co-formulation, co-implementation of SD). Nevertheless, you will have the opportunity to practice individual design skills during Module 3, where the initial phases of the design process are done individually, until the midterm presentation. The midterm moment includes two types of formative feedback: from coaches and clients during the presentation; and from student peer feedback. From this point onwards, students will choose a direction together from the individual contributions and continue working in teams for the remainder of the assignment of module 3.	
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Special Information	Contact: stratds-ide@tudelft.nl	

IDEM3101	Foresight and Backcasting	5
Course Coordinator	Dr.ir. W.L. Simonse	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This is an SPD programme core course. Time is of strategic importance in understanding, leveraging and influencing socio-technical transitions. Recent research in the field of strategic design indicates an important shift from anticipating and coping with the future towards future making and designing foresights. This course takes a particular interest in collectively designing and imagining foresights by expressing and materializing future visions in strategic practice. In this course you will learn and practice methods and techniques of creative trend research, designing future visions and back-casting by a strategic design roadmap. You will develop analytical, envisioning, and modelling skills vis-a-vis providing strategic support for organizations to innovate along meaningful pathways (future literacy, exploratory mindset & imagination, and responsible design & engagement).	
Study Goals	Upon finishing this course, students are able to: 1. Analyze socio-technical transitions, strategies, purposes, and priorities of organizations and communities. 2. Create trend foresight research to uncover future use values in user- and design communities. 3. Envision new grounds by modelling future values, imaginative time travels, and expressing collective conversations about desired futures. 4. Make a future vision artefact to support strategic exploration and eco-system communication. 5. Back cast from a future vision to an existing value proposition by a horizons time pacing of a strategic design roadmap.	
Education Method	Experimental learning in real-life projects on designing foresights based on two learning cycles: I. Trend Foresight - Reading trend foresight literature - Design research by each student II. Future Making - Reading future visioning + back-casting literature - Designing and making a future vision artefact - Design a strategic roadmap.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4337 Design Roadmapping.	
Assessment	The study goals of this course will be assessed by: Real life project-based deliverables: - Creative trend foresight research (50%, individual assignment) - Future making artifact + strategic design roadmap (50%, group assignment)	
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Special Information	Contact: fab-ide@tudelft.nl	
Remarks		

IDEM3102	Research Methods for Strategic Design	5
Course Coordinator	Dr. E. Kalmár	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This is an SPD programme core course. The aim of this course is to introduce the fundamentals of robust, reliable, and valid social scientific research related to strategic design and guide students through the process from topic selection to data presentation. The students will learn the differences between research for/about/through design, how to use theories in research, create strong research questions, choose a fitting research design, define participants, design reliable research instruments, and finally collect, analyze and present data. They will practice the learned material by concrete assignments. The course will cover qualitative and quantitative academic research methods for the two key phases of innovation in which strategic designers operate, namely the fuzzy-front end and muddy back-end. In particular, the qualitative part will cover interview techniques, focus groups and ethnography applied to research topics relevant to strategic design research projects (e.g., drivers of smart products adoption, organisational and personal drivers in sustainable transitions, understanding stakeholders' needs in healthcare ecosystems), while the quantitative part will focus on building reliable surveys for research topics relevant to the muddy back-end (e.g., market introduction of smart projects, concept evaluation, tool probing). The course will put particular emphasis on creating solid arguments through research data by taking validity, reliability, generalizability and research ethics into account. Guest lecturers will present their strategic design-related research projects from the perspective of the weekly topics.	
Study Goals	Upon finishing this course, students are able to: 1. Create a research plan and justify the methodological decisions by describing the importance, benefits, and limitations of different research designs the context of strategic design. 2. Identify valid, ethical, and reliable research methods and incorporate these issues into the research design. 3. Use theories in the various phases of the research process. 4. Apply basic quantitative and qualitative methods to the given research context and perform a basic analysis of qualitative and quantitative research data.	
Education Method	Lecture (2h) and practical part (working on assignments with TAs) (2h).	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4355-18 SPD Research.	
Assessment	The study goals of this course will be assessed by: - Individual assignment (50%): a detailed research proposal containing a research question, a research design, methods choices including sample, instruments, analysis plan and reasoning supporting the choices - Individual written exam (50%)	
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Special Information	Contact: rmsd-ide@tudelft.nl	

IDEM3103	Deep Dive into Organisations	5
Course Coordinator	Dr. S.N. Nikou	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This is an SPD programme core course. In this course students will work with theories and cases that will give them advanced knowledge about organisations and their surrounding e.g. stakeholder network ecosystem, how they innovate, and the way in which strategic designers can interact with and add value to them. The first part of the course will focus on different organisational theories and why different kinds of organisations exist and what are their innovation value drivers that can influence and be influenced by strategic designers. Organisational capabilities such as innovation portfolio, differentiation from competition, financial and business model sustainability will be covered. The 2nd part of the course will take a more systemic perspective and will look at organisations as parts of ecosystems. Students will learn how organisations connect, relate and compete in the context of socio-technical and/or SDG-driven transitions, as well as how strategic design capabilities can play a role in these transitions (organisational transformations that occur within them).	
Study Goals	Upon finishing this course, students are able to: 1. Students are able to summarise and compare current developments in organisational theories and theoretical advancements related to the strategic value of design skills and capabilities. 2. Students are able to illustrate and evaluate the role and relationships of different kinds of organisations in SDG-driven transitions. 3. Students are able to identify and analyse the value drivers of organisations and ecosystems. 4. Students are able to develop, articulate, and defend well-founded opinions on the role of strategic design within and across organisations.	
Education Method	This course follows a combination of in-class (guest) lectures, real-world case studies, and group discussion. The group discussions will use two formats: the case method (for real case studies) and the statement-based discussion with experts (for the sessions with guest lecturers).	
Assessment	The study goals of this course will be assessed by: - Individual written final exam (60%) - Group assignment on a case, short report (40%) Group presentation will be assessed as a feedback moment (formative assignment).	
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Special Information	Contact: ddio-ide@tudelft.nl	

Year	2024/2025
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SPD Program Electives (minimum of 15 EC)

IDEM3201	Qualitative Research Methods for Strategic Design	5
Course Coordinator	Dr. S. Hiemstra-van Mastrigt	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective related to SPD themes 1 and 2. This course teaches students qualitative research techniques in relation to Strategic Design. This includes interviews, focus groups, case study methodology, action research, ethnography. Students practise with applying the methodologies to a real research topic as proposed by researchers in our faculty. Students can choose a topic. The research topics will be related to one of the two themes (Tech-enabled innovation or Social and Sustainable Innovation), so that students can keep deepening within their knowledge area of interest. The course supports students in applying and understanding scientific research methods so they can (1) successfully perform qualitative research projects by themselves, and (2) critically evaluate the relevance and value of outcomes of qualitative research. This course will teach students how to design a research plan, how to collect data, how to analyse data and how to communicate the research results.	
Study Goals	Upon finishing this course, students are able to: 1. Evaluate different qualitative research designs, methods and techniques for strategic design 2. Perform qualitative data collection and analysis 3. Evaluate and improve the quality of qualitative research 4. Communicate research outcomes	
Education Method	Lectures, tutorials and group work exercises, coaching	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4355-18 SPD Research.	
Assessment	The study goals of this course will be assessed by project group work (100%) In the report, the research project of the group will be presented as a scientific article, including prior literature, research method, results, and discussion and conclusion sections.	
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Special Information	Contact: qual-ide@tudelft.nl	
Remarks	Information for this course is only available in English. Select the English language in the top right corner.	

IDEM3202	Quantitative Research Methods for Strategic Design	5
Course Coordinator	G. Granato	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Expected prior knowledge	The course IDEM 3102 represents the preliminary course that we recommend to have followed before taking this course. If students did not follow the course IDEM 3102, we highly recommend to study the course material of IDEM 3102 at least (mainly the quantitative part- knowledge on dependent and independent variable, hypothesis formulation, basic knowledge of SPSS, t-test, etc..).	
Course Contents	This is a programme elective related to SPD themes 1 and 2. This course teaches students quantitative research techniques in relation to Strategic Design. This includes experiments, hypothesis testing, and statistical analysis techniques. Students practice with the learnt methodologies through tutorials and feedback hours. The course supports students in applying and understanding scientific research methods so they can (1) successfully perform quantitative research projects and data collection and (2) critically evaluate the relevance and value of quantitative research outcomes.	
Study Goals	Upon finishing this course, students are able to: 1. Evaluate different quantitative research techniques and analysis methods for the field of strategic design (This entails identifying their strengths and weaknesses, and their suitability for investigating different types of research questions) 2. Analyse the impact that a research has on user responses (cause- effect relations) in the field of strategic design. 3. Design and conduct a quantitative research (experimental research) in the field of strategic design and analyse its data. 4. Report and discuss the research findings of a strategic design research following the scientific standards.	
Education Method	Lectures, tutorials and group coaching.	
Computer Use	All lectures and tutorials will take place in the computer room. Students will need their own computer with SPSS programme downloaded for all the tutorials and group works.	
Literature and Study Materials	Slides, book and tutorials represent the study material.	
Books	Students are asked to study chapters of the book "Research Methods in human-computer interaction", second edition by J. Lazar, J.H. Feng and H. Hochheiser. This book is also used in another IDE course, therefore can be found second-hand (as well as new and online). This book represents the study material for this course exam, together with other material specified in the lectures.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4355-18 SPD Research.	
Assessment	The study goals of this course will be assessed by: - Individual exam (40%) - Group report (60%) In the exam, general knowledge on quantitative research methods and data analysis will be individually assessed. In the report, the research project of the group work will be presented as a scientific article, including prior literature, research method, results and discussion.	
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Special Information	Contact: quant-ide@tudelft.nl	

IDEM3211	Technology and Strategic Design	5
Course Coordinator	Dr. V.P. Pannunzio	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	This is a programme elective of SPD theme 1: Strategic Design for Technology-enabled Innovation. This course is designed to empower you with foundational theoretical insights into the trajectories of technological innovation. This course is your gateway to becoming a strategic designer that drives technology-enabled innovation within organizations and across ecosystems. You will delve into the evolution of technologies, their integration with innovation management and strategy, and their alignment or misalignment with societal needs and the rapid pace of change. Explore cutting-edge technologies such as the Internet of Things (IoT), brain decoding, artificial intelligence, and blockchain, understanding their implications and potential. This course is tailored to foster your autonomy in acquiring essential knowledge of emerging technologies. Engage with technology experts, confront key ethical and ecological dilemmas inherent in technological advancements, and ponder the role of strategic design in crafting solutions that yield a positive impact on society. Prepare to challenge your perspectives and contribute meaningfully to the discourse on technology and innovation.	
Study Goals	Upon finishing this course, students are able to: 1. Investigate existing and emerging technologies from a (strategic) design perspective. 2. Explain the role of strategic design in future technological innovation trajectories. 3. Discuss ethical, environmental and societal implications of technology development and application. 4. Present and uphold arguments to peers and technology experts through professional-level oral presentations and reports.	
Education Method	The course will be taught through a mix of interactive lectures (including guest lectures with experts), plenary discussions and debates, workshops, and recommended materials.	
Assessment	The study goals of this course will be assessed by: - Assessment 1 (report, group) graded 1-10 (100%) - Assessment 2 (presentation, group) - pass/fail* (*the calculation of the final grade (which is graded 1-10), but a pass grade in exam 2 is necessary for passing the course.) You will demonstrate your mastery and understanding of the technology in a group presentation, set against the backdrop of an audience comprising technology experts, your peers, and the course instructors. This presentation is your platform to demonstrate the depth of your knowledge on a selected emerging technology, applying the insights and concepts learned throughout the course. In this presentation, you will be asked to: - Elaborate on the key principles of the emerging technology, its evolution and its future potential; - Articulate the role of strategic design in relation to the emerging technology; - Reflect on the emerging technologies ethical, environmental and societal implications; Prior to the final presentation, you will have the opportunity to refine your understanding and approach through a formative assessment. This will involve a peer- and self-review of an early version of your presentation, allowing for constructive feedback and insightful reflection to guide your improvements. Both the formative and summative assessment of the presentations will be conducted using a standardized rubric.	
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Special Information	Contact: tsd-ide@tudelft.nl	

IDEM3212	Tech-enabled Entrepreneurship	5
Course Coordinator	Ir. R.G.H. Bluemink	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of SPD theme 1: Strategic Design for Technology-enabled Innovation. This course focuses on intra- and entrepreneurial aspects of technology-based innovation, from creating and validating user-driven value propositions, developing their related business models, and assessing financial viability. Students will learn how to apply business economics theory, practices and tools to conceptualise and organise tech-enabled innovation ecosystems in a multi-actor context. In this course, we teach how to finance the development phase of a new value proposition, by finding the appropriate business model, by attracting external financing and by optimising internal financial management. Students frame financial viability from a business model perspective, shifting their focus from generating value to delivering and capturing value for ecosystem actors. The course covers product-service cost structures, sales revenue planning, capital needs planning, income statement, cash flow planning, break-even analysis, and financial risk assessment. Finally, students will learn how to balance financial goals with ecological and societal considerations and constraints. Therefore, sustainable business models and the triple bottom line are also covered in this course.	
Study Goals	Upon finishing this course, students are able to: 1. Identify and formulate meaningful user-driven product-service value propositions for tech-enabled innovations. 2. Set up a tech-enabled innovation ecosystem considering stakeholder interests in a multi-actor context. 3. Calculate a financial business case for a user-driven product-service value proposition. 4. Evaluate financial, environmental, ethical and social values for tech-driven initiatives.	
Education Method	Lectures (theory and methods) combined with sub assignments (formatively peer-assessed to minimise staff efforts). Guest lecturers from practitioners & specialists.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4330 New Product Economics.	
Assessment	The study goals of this course will be assessed by a summative assessment of a group assignment (100%). The assignment is based on a real case or a fictitious case. Students (4 per group) set up a business case for an innovation ecosystem (fictitious). The group assignment has one individual component (peer assessment).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: tee-ide@tudelft.nl	

IDEM3213	Brand Experience Across Technologies	5
Course Coordinator	Ir. S. Wu	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This is a programme elective of SPD theme 1: Strategic Design for Technology-enabled Innovation Branding can support organisations in creating acceptance for tech-enabled innovations and in smoothly integrating digital technologies in current offerings. Designers can play a central role in this through brand experience design, namely by creating meaningful and coherent experiences across digital and physical touchpoints and by keeping the experience aligned with the core values of an organisation. In this course students will learn different brand frameworks and how they can be used to create a branding strategy that supports tech-enabled innovation. They will then learn how to translate a brand strategy into design guidelines and design manifestations that convey the brand identity in a consistent and authentic manner. They will discover how to use design practices like co-creation, empathy, and prototyping to deliver rich brand experiences across different types of touchpoints.	
Study Goals	Upon finishing this course, students are able to: 1. Understand several brand models, apply one of these models to analyse an existing brand. 2. Analyse product, organisation, competitors, consumer and context (4Cs) in the light of brand development 3. Integrate the outcomes of analyses to create a brand strategy for new and/or existing brands. 4. Translate brand strategies into design guidelines for authentic and coherent brand experiences across digital and physical touchpoints and evaluate their effectiveness in the design process. 5. Create and evaluate meaningful brand experiences based on the brand strategy.	
Education Method	Students will receive blended learning activities including the following elements: 1. (guest)lecture to explain and discuss the key learnings (average 1 hour/week). 2. studio-based coaching sessions (average 1 hour/week): coaches provide guidance on project planning, critical insights on decisions and formative feedback. In wk 4 and wk 8, students present their intermediate results and receive coach and peer feedback. 3. literature self-study: 2 hours 4. group assignment (in the form of a project): to apply knowledge and insights. It consists 2-4 hours of individual work (tasks assigned by the team); and 6-8 hours of group work (planning, decisions, idea generation, selection; concept evaluation).	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4315-16 Brand & Product Commercialisation.	
Assessment	The study goals of this course will be assessed by a group assignment grade adjusted by the individual contribution through peer evaluation (100%). The following principles apply to peer evaluation: - Anonymous assessment of each others contribution - Based on fairness and facts - Monitored by the coaches	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: beat-ide@tudelft.nl	

IDEM3221	Societal Missions: Theory and Practice		5
Course Coordinator	W. Brouwer		
Contact Hours / Week x/x/x/x	4		
Education Period	3		
Start Education	3		
Course Contents	This is a programme elective of SPD theme 2: Strategic Design for Social and Sustainable Innovation. To equip yourself as future strategic designer working on societal missions, this course will provide you with a solid theoretical knowledge base on societal missions. By evaluating this theoretical knowledge you will better understand dynamics at work in social missions. To help you analyse these dynamics in reality, you will be confronted with practitioners working on societal missions. In this course you will dive into different theories on political and organizational dynamics in societal missions, different levels of power and relationships, theories on working with differences scales (micro/individual-meso/organization-macro/system level) and implementing missions on different time-frames (short and long-term). You will learn from dialogues with practitioners working on societal missions, for example in, energy transitions, healthcare, and, a just society from different types of organizations. And you will analyse the practice in reality by having a conversation with practitioners on their work. The course is individually assessed, however you will team up with a buddy to prepare your conversations together and also observe each others conversations. By doing so you develop a good understanding of what it means to work on societal missions as a strategic designer.		
Study Goals	Upon finishing this course, students are able to: 1. Compare and contrast theories on societal missions in order to evaluate their suitability to design practice. 2. Critique a practitioners account of their design practice based on societal mission theories.		
Education Method	Reading literature, workshops, practitioner talks or site visits.		
Assessment	The study goals of this course will be assessed by an individual assignment (100%). You will be assessed individually. Formative assessment will take place during the course through dialogue sessions and peer feedback. Moreover, you will do 1 practice conversation with a practitioner to practice their final assignment. You will get plenary feedback on this formative assignment from teachers as well as direct peer feedback.		
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.		
Special Information	Contact: smtp-ide@tudelft.nl		

IDEM3222	Responsible Marketing	5
Responsible Instructor	B.R. Baldassarre	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of SPD theme 2: Strategic Design for Social and Sustainable Innovation. In responsible marketing, students will learn how organizations can facilitate adoption of "meaningful and sustainable" solutions that can help in the move to a circular society. Segmentation based on benefits sought and jobs-to-be done will be essential to target early adopters that can influence the mass-market via word-of-mouth (offline and online). The course will discuss sustainability/performance trade-offs in the context of responsible solutions, and how marketing strategies and tactics can mitigate these trade-offs. Students will also learn how companies are moving from green marketing to sufficiency marketing, in which the sole focus is not on selling more but enough products. For true transitions to occur, the course will also tackle the roles of multiple parties in the value chain (think retailers and food/packaging waste) and in the innovation ecosystem (think electric boating in the Amsterdam canals).	
Study Goals	Upon finishing this course, students are able to: 1. Explain and apply segmentation, targeting and positioning to facilitate responsible marketing strategies 2. Explain and analyse how organizations can leverage the marketing mix to facilitate the adoption of sustainable innovations 3. Apply design practices to integrate different stakeholders perspectives in the sustainable solution 4. Apply and analyse responsible marketing knowledge with hands-on experience via a circular marketing simulation to develop marketing strategies that meet profit, people and planet objectives.	
Education Method	First 4 weeks: 2-hour lectures Last 4 weeks: Circular MarkStrat simulation, 2 rounds per week, total 8 rounds	
Assessment	The study goals of this course will be assessed by: - Individual essay (50%) - Presentation on Circular MarkStrat results (50%) Individual essay on analyzing and applying the responsible marketing strategies of a chosen organization (based on articles and lecture material): 50%. Individual peer review of essay after 3 weeks (feedback moment: formative assignment). Group presentation on Circular MarkStrat team results and strategies with individual questions: 50%. Students play a simulation game in 8 rounds; in each round they need to take multiple responsible marketing decisions. At the end they need to present their final results of their products in the market and the effectiveness of their responsible marketing strategies. Each individual team member will receive questions about the decisions taken.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: rm-ide@tudelft.nl	

IDEM3223	Sustainable Consumer Behaviour	5
Course Coordinator	Dr. L.B.M. Magnier	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This is a programme elective of SPD theme 2: Strategic Design for Social and Sustainable Innovation. Present consumption patterns have destructive effects on eco-systems. It is designers responsibility to tackle these negative consequences and design products, services and systems that contribute to a sustainable society. However, acceptance and adoption of these innovations / interventions by consumers is not always straightforward, which may ultimately reduce their positive impact on society. In this course, we will touch upon a number of individual and systemic factors that are known to be of high relevance for sustainable behaviour change and that are relevant for the success of a Circular Economy. Furthermore, we will explore how various interventions can stimulate more sustainable consumer behaviour. Topics that will be addressed are: How do consumers develop attitudes towards sustainable innovations and make choices between products? How do consumers need to change their behaviours in a Circular Economy and how can design contribute to these changes? What is the value of product design for consumers perception towards sustainable products? How can design contribute to breaking existing habits and forming new sustainable ones?	
Study Goals	Upon finishing this course, students are able to: 1. Explain the most relevant theories relating to sustainable consumer behaviour. 2. Independently conduct a scientific literature review on the topic of sustainable consumer behaviour. 3. Apply theories relating to sustainable consumer behaviour to form an analysis and predict responses to an innovation/intervention. 4. Integrate various theories of sustainable consumer behaviour in a convincing and clear textual essay 5. Formulate and visualise recommendations about an innovation/intervention based on their analysis of the literature in sustainable consumer behaviour	
Education Method	Lectures + group project (essay) Lectures are on-campus. Students should meet with each other to write their essay and organise coaching sessions with their coach at a time that fits both the coach and the group. They send their essay at least 2 days before the coaching session so that their coach has the time to review it and provide comments during the meeting.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5136 Sustainable Consumer Behaviour.	
Assessment	The study goals of this course will be assessed by: - Group assignment (50%) - Individual Exam (50%) The group assignment consists of an essay where students are asked to analyse how consumers are likely to respond to a recent innovation /intervention basing their arguments on the scientific literature in sustainable consumer behaviour. Based on this analysis, they are asked to provide visualised recommendations on how to improve the concept. Students receive formative feedback on their essay during the coaching sessions.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: scb-ide@tudelft.nl	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

Design Studio Spring Semester (minimum of 10 EC)

IDEM3210	Tech-enabled Innovation Studio	10
Course Coordinator	J. Jung	
Contact Hours / Week x/x/x/x	8	
Education Period	3 4	
Start Education	3	
Course Contents	This is a programme elective design studio of SPD theme 1: Strategic Design for Technology-enabled Innovation. Do you fancy technology? Nice, because nowadays, innovation is frequently centred around emerging technologies. Strategic design can play a central role in making sense of those emerging technologies and helping organisations and ecosystems apply those in their innovation process. In this hands-on course students go through all the steps and challenges of a strategic design project in the context of tech-enabled innovation. You will work on real client projects focused on different technologies and on different application areas (healthcare, mobility, energy transition, finance etc), spanning from technology exploration to developing tech-enabled value propositions. You will learn how to adapt key strategic design capabilities to this specific context and how to integrate the use of tech-enabled prototypes and experiments in the different phases of the strategic design process. You will bring together the different perspectives needed to solve the puzzle by setting up multi-actor collaborations (i.e. designers, developers, data-analysts, product owners etc) around the project. Data is a great basis on which to motivate and justify the integration of technology into the design process and the product. Thus, the course will also zoom in on use of data in a strategic context, by looking at how thick/thin and big/small data can be used as an input throughout the strategic design process. You will learn how data can be used as a means of sense-making, analysis, presentation, and evaluation to facilitate strategic decision-making. And while innovating, you will reflect on the ethical and legal aspects of the applied technology, such as GDPR and inclusivity perspective. At the end of the course, students will present their work to real-life stakeholders in a presentation and a report.	
Study Goals	Upon finishing this course, students are able to: 1. Develop society-challenge-solving tech-enabled innovation outcomes. 2. Apply iterative methodologies and design-driven practices in the strategic design process both in problem framing and solution developing phases. 3. Identify different disciplines and stakeholder related to their project and analyse their own role in the overall design process. 4. Evaluate the integration of tech-enabled practices in the strategic design process and for strategic decision-making through analysis of thick/thin and big/small data. 5. Present your work with the logics of your decisions in the work as a form of a live presentation and a written text report. 6. Reflect on the ethical and legal implications and perspectives of the applied technology 7. Evaluate the characteristics and impact of the chosen technology	
Education Method	Project-based in groups of 5 students interacting with their clients on a regular basis, being coached by a coach from TU Delft. Integration of individual and group design outcome developments, being coached by a coach from TU Delft and inputs from external clients (Firstly ask to develop individual design outputs and make an evaluation as a group to decide that groups direction). Guest-expert lectures and workshops on skills to present their works (presentation and report), and iterative methodologies. Our contact hours will varied 4-8 hours. Some weeks will involve workshops that require longer contact hours. In general, Coaches would be available to talk to students for 4 hours per week. Students will also have the opportunity to pitch new technologies and related opportunities. Their pitches will allow other students to learn about these technologies. Resident experts will also provide feedback on these pitches to ensure accuracy of representation. While they are taking this course, many students also learn about the technologies to be used in design processes from other tech-enabled track electives (e.g. Tech-enabled entrepreneurship & Technology and Strategic design courses), and this opportunity allows them to share what they have learned from the other electives.	
Assessment	The study goals of this course will be assessed by: Summative assessment (pass/fail) (100%) - Group assessment: Project presentation to client, assessed by the coach using a rubric. - Group assessment: Design report to present group project process and outcomes, assessed by the coach using a rubric. - Individual assessment: Written report on the students individual role in the project, assessed by the coach using a rubric.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: techds-ide@tudelft.nl	
Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.	

IDEM4220	Systemic Design Studio	10
Course Coordinator	G.S. Celik	
Contact Hours / Week x/x/x/x	8	
Education Period	3 4	
Start Education	3	
Course Contents	This is a programme elective design studio of DfI theme 2: Design for Societal Innovation and SPD theme 2: Strategic Design for Social and Sustainable Innovation. In this immersive design studio, you will experience designing in the context of complex societal challenges in a multidimensional way and learn how to adapt key capabilities to this specific context. You will engage with real-world challenges and learn how to create design interventions (artifacts, services, strategies, processes or any other manifestations of design) that intentionally contribute to system-level change. You will do this by envisioning alternative futures and designing for them through the use of systemic design methods and principles. You will learn how to perceive societal challenges and the way to address them through various theoretical lenses including complexity theory, network theory and evolutionary design. You will also learn research methods to investigate complex societal challenges, as well as systemic design principles to design for them. The theoretical and methodological knowledge will be integrated in the design project that you will undertake as a part of this hands-on course. Throughout the course you will be working on multiple real challenges coming from clients in consecutive design cycles. Design skills and capabilities like (visual) communication techniques for systemic design and design leadership in public/private ecosystems will also be practiced in this course. Finally, while expanding your portfolio through systemic design, you will also have the opportunity to expand your network through engaging with stakeholders in relation to your chosen challenges and conduct field research as you follow participatory design approaches to learn how to balance the interests and concerns of individuals, communities, organizations and other actors that form the society.	
Study Goals	Upon finishing this course, students are able to: 1. Critically discuss the evolving field of systemic design and designers role. 2. Apply systemic research methods and a complexity lens to analyze societal challenges. 3. Communicate the complexity of a societal challenge in a way that is understandable, compelling and actionable. 4. Integrate systemic principles in designing interventions and their reasoning. 5. Develop and execute an approach for effectively engaging system stakeholders that is justified and adapted to the design context. 6. Conduct safe-to-fail experiments that result in systemic interventions accompanied with prototypes. 7. Formulate a future vision and a pathway (including portfolio) towards systemic change. 8. Anticipate and propose means and actions for shaping alternative futures by systemic design.	
Education Method	The course will primarily follow a studio structure with project groups of 4-5 students working on the same challenge. Additionally, there will also be an individual design assignment, where students get to excel in individual design competences as a preparation for their graduation project. A combination of (guest) lectures, field visits, workshops and studio coaching will be used as forms of instruction throughout the course. Coaching hours vary between 4-8 hours per week depending on the program. Each team (or student when working on individual cases) will be guided by a TU Delft-based coach whose expertise is in systemic design, strengthened by the input of the client/challenge owner. The studios will receive visits from clients/challenge owners, during which students can pitch ideas, receive information or organize participatory sessions. In this course IDE MSc students from both SPD and DFI programs will work together in project groups to complement each others skillsets and approaches to complex design challenges from multiple perspectives. In order to gain experience with different paces, levels of depth and client briefs, the students will get the chance to work on multiple design cycles that follow each other. At the end, students will report on their process and also present their work to each other in an interactive manner during a project market, where they get to address wider audiences in an informal set-up.	
Assessment	The study goals of this course will be assessed by three summative assessment moments (resulting in Pass, Fail or Fail with opportunity to compensate), at the end of each design cycle. To be able to pass the course, students need to receive a pass on all three assignments. Each design cycle emphasizes different parts of the systemic design process and allows the students to dive deeper in exploration, communication, intervention and reflection. The assessments will be done through a pass/fail model to support experimentations with various systemic design principles in the course. While continuous coaching will ensure expert feedback, the course will also involve multiple peer feedback moments where the students can share progress and engage with the rubric at the same time. The distribution between group and individual work across the three summative assessments is as follows: Assessment cycle 1 (group) design a systemic communication intervention (presentation) For A1 student groups will be asked to present the outcome of their design work and describe their design process for the summative assessment (pass/fail). There will be a mid-way presentation as well, where students receive formative feedback on their progress (based (on rubric). Assessment cycle 2 (individual): design systemic intervention concept (video presentation) At the end of cycle 2 the students will be asked to submit a short video presentation of their conceptual design, which forms the summative assessment (pass/fail). Continuous guidance will ensure coach feedback and students will also receive peer feedback in an organized session. Assessment cycle 3 (group with an individual reflection component): detail systemic design intervention and evaluation (report and presentation) In A3 students will work in groups again to further iterate on (one of their chosen) design concepts. They will first present their detailed design work to peers, coaches and clients and receive formative feedback (based on rubric). Later on, the groups will be asked to submit a short written document that describes the outcome and the process, which forms the summative assessment. A part of this document will be dedicated to individually written reflections. The final grade (pass/fail) will be calculated individually. The course has a strong emphasis on group work, where students work in a group during 15 of the 20 course weeks to reflect the collaborative nature of systemic design practices. However, we want to highlight the importance of gaining creative confidence individually when designing for systemic change and therefore the second design cycle is formulated as an individual assignment. Despite suggesting an individual journey, the second design cycle will benefit from the interactive atmosphere in the studios and the process will be strengthened by peer feedback in addition to coach and client support. After the individual part, students will come back together in groups and collectively choose a path to continue.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details:	

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Special Information

Contact: sysds-ide@tudelft.nl

Year	2024/2025
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

2nd year Master Strategic Product Design

Year	2024/2025
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

Electives (minimum of 30 EC)

IDEM101	Advanced Service Thinking and Design Practice	5
Course Coordinator	Dr. H.M.J.J. Snelders	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	The last decades have seen the application of already existing design practices to various service industries, with some degree of success. In this elective, we will explore how design can advance further in the service sector through a more specialised mindset and skill set for design research, ideation, and prototyping. The course is composed of two main parts: Part 1, on service design research, highlights how service users are always strongly connected to the production of a service. Special attention can be paid to this in design research, and we will offer hands-on exercises in methodologies that help you do so (notably Service Safaris and Journey Mapping). Part 2, on new service ideation and prototyping, highlights how value creation in services relies on collaborations between service users and producers that can only be partially controlled and planned for. Here as well we will offer hands-on exercises, using specialised methods (notably Bodystorming). Furthermore, the elective will work with real life cases from multiple service sectors (e.g. health, mobility, information, legal). Most exercises are group work, set up as a codesign effort where case interests will be weighed against your personal and group learning goals for developing your mindset and skill set for service design.	
Study Goals	Upon finishing this course, students are able to: 1. produce visual documentation of experiencing a current service that is creative and insightful, by using a specific service design tool. 2. Analyse and synthesise visual documentation of a current service that is creative and insightful of case information, by using a specific service design tool. 3. Co-create new service propositions that are relevant within the case, by using specific service design tools. 4. Demonstrate how the results from service design ideation and prototyping stem from a balanced dialogue between stakeholder interests and personal and group learning goals for competency development in service design.	
Education Method	Two types of learning activities are organised: 1) Through lectures, reading, and workshops students derive personal (yet theory-informed) learning goals for competence development in service design. These activities will take place in the first two weeks and middle of the course 2) Through real, case-based exercises with service design methods students can apply their learning goals in co-creation with case stakeholders. Next to a small individual first week exercise, these activities will take place from week 3 onwards.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5666 Service Design.	
Assessment	The study goals of this course will be assessed by: - Group work (80%) - Individual assignment (20%) All learning objectives will be assessed through formative feedback (throughout the course), and a summative review and grade (in week 4 and week 9/10).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: serdes-ide@tudelft.nl	

IDEM102	Build Your Start-up 1: Idea Validation	5
Course Coordinator	Ir. M.J.J. Buijs	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	The BYS 1 + 2 elective teaches how to build and validate the fundamental building blocks of a startup. Students start with a venture idea that slowly gets morphed into an actual startup with paying customers. Its one of the few, if not the only, course where students actually get to implement their own solution in the market. BYS 1 focusses on idea development and validation with actual customers, industry experts and other external stakeholders by applying methods such as problem interviewing, agile design & lean startup principles to assess desirability and first hints of feasibility and viability. Key question: is the venture idea promising enough to operationalize it in Q2?	
Study Goals	Upon finishing this course, students are able to: 1. Justify their value proposition and initial customer segment using generated evidence 2. Generate evidence by means of experiments and interviews with potential customers 3. Identify and make explicit the risky assumptions and weaknesses in their startup in a validation board 4. Reflect on their actions as an entrepreneur by writing an individual reflection paper 5. Explain the potential value of their startup to potential investors by crafting and doing an investor pitch	
Education Method	In BYS courses, the emphasis is put on individual team coaching sessions, peer feedback and events where experienced entrepreneurs and experts (often course alumni) guide students pro-bono to validate their assumptions and build their idea. A mix of workshops and lectures is aimed to inspire and guide the students and forms a string of building blocks towards validating (BYS1) and operationalizing (BYS2) the venture idea. BYS courses are not ABOUT entrepreneurship. BYS courses are about DOING entrepreneurship. At BYS we tailor the journey around what your team needs at the very moment. We'll help students to Build Your Startup, one step at a time.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5659-1 Build Your Startup 1: Idea Validation.	
Assessment	The study goals of this course will be assessed by: - A Pitch of the startup and validation evidence (team assessment) (pass/fail) - A personal reflection using the EnJournal method (individual assessment) (pass/fail)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: bys-ide@tudelft.nl	

IDEM103	Build Your Start-up 2: Prototyping	5
Course Coordinator	Ir. M.J.J. Buijs	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	The BYS 1 + 2 elective teaches how to build and validate the fundamental building blocks of a startup. Students start with a venture idea that slowly gets morphed into an actual startup with paying customers. Its one of the few, if not the only, course where students actually get to implement their own solution in the market. BYS 2 focusses on operationalizing the venture idea by forecasting and experimenting to validate: Market need. Getting actually first paying customers, Go2Market strategy, marketing campaigns Technology. Creating MVPs Operations. Start operating as a real business, with actual roles and goals, setting up the supply chain Setting up prices. Key question: is there initial product-market fit that proves the venture could become a success?	
Study Goals	Upon finishing this course, students are able to: 1. Justify their value proposition and initial customer segment by (mockup or real) sales. 2. Define a project plan for further operationalisation of the startup after graduation. 3. Design & conduct experiments to research desirability, feasibility and viability of the venture. 4. Reflect on their progress as an entrepreneur & willingness to continue the journey & personal plan of action for the future by writing an individual reflection paper. 5. Show the progress of the startup & prove you are the team to do it to potential investors by crafting and doing an investor pitch.	
Education Method	In BYS courses, the emphasis is put on individual team coaching sessions, peer feedback and events where experienced entrepreneurs and experts (often course alumni) guide students pro-bono to validate their assumptions and build their idea. A mix of workshops and lectures is aimed to inspire and guide the students and forms a string of building blocks towards validating (BYS1) and operationalizing (BYS2) the venture idea. BYS courses are not ABOUT entrepreneurship. BYS courses are about DOING entrepreneurship. At BYS we tailor the journey around what your team needs at the very moment. We'll help students to Build Your Startup, one step at a time.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5659-2 Build Your Startup 2: Prototyping.	
Assessment	The study goals of this course will be assessed by: - A Pitch of the startup, plan for further operationalisation and validation evidence (team assessment) (pass/fail) - A personal reflection using the EnJournal method (pass/fail)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: bys-ide@tudelft.nl	

IDEM104	Creative Facilitation	5
Course Coordinator	K.G. Heijne	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	The Creative Facilitation course offers an in-depth experiential learning laboratory in the use of creativity techniques and the facilitation of creative processes. Facilitation in this context relates to someone whose role it is to work with a diverse group of people to ensure that a creative journey/process leads to innovative and implementable results with a high degree of consensus. The course starts with building a shared science-based vocabulary through literature and guided exercises and then moves on to hands-on experience through experiments and cases, each followed by presentation of results and reflection, in other words: experiential learning and reflective practice. During the course, we will look at the creative person, explore a wide range of creativity principles and techniques and develop methods to design processes for innovation teams. Creativity is not an end in itself, it is always applied to something; therefore, we will invite real clients with real problems: ranging from design challenges to new business opportunities to problem solving.	
Study Goals	By the end of the course, students will have an extended experience of organizing, facilitating and participating in creative processes. The actual outcome will vary for each student and depends largely on course objectives as set by each individual student. The course is set up in such a way as to provoke a reflective conversation on the creative process in small groups on the basis of theory and hands-on experience. Upon finishing this course, students are able to: 1. Apply the basic principles of facilitating creativity in group sessions. (Apply) 2. Compose a session plan and select appropriate creativity techniques. (Create) 3. Appraise the quality of creative sessions, by recognizing strong points and points for improvement in creative sessions the student attends either as facilitator or resource group member. (Evaluate) 4. Communicate clearly and interact professionally with the problem owner prior, during and after the creative session. (Apply)	
Education Method	Week 1-6: The first 6 weeks of the course we will apply the flipped classroom concept. The students will read theory and watch videos on the basic principles of facilitating creativity in groups. They will also make assignments individually. During the contact hours, the focus will be on guided exercises which gradually grow in complexity and scope. This is followed by sessions organized by students, guiding a group of their peer students from the course. In these sessions the students will experience a wide array of techniques and gain various insights in facilitation, group dynamics, problem analysis, idea generation and selection, and implementation. Each experiment is followed by an extensive (peer) reflection. Week 7-10 will be based on intensive experiential learning: setting up and leading sessions on all kinds of issues, ranging from design challenges to new business opportunities to problem solving. These sessions involve real issues from real clients. Each experience is followed by an extensive reflective conversation. The students will be encouraged to keep a journal to capture all learnings. This journal should also help them to get quicker into the right mindset necessary for this course. NOTE: Due to the steep learning curve, the hands-on approach and the group work, you will gain the best learning experience if you attend ALL days.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5325 Creative Facilitation.	
Assessment	This course was designed to focus on learning, understanding, and mastering the skills for facilitating creativity in groups, inspired by Beghetto (2005). To create optimal creative press, any creative output in this course will be assessed formatively only. The summative assessments are focusing on whether students will take the chance to experiment, how well they understand the principles and vocabulary, and how they capture what they have learned from their own and their peers experiments. Formative assignments on which the students will share and receive peer evaluation are: Learning contract (individual): an artefact, representing your personal learning objectives for this course and your commitment to achieve this. Practice sessions and creativity techniques (in duos): various rounds of sessions will be facilitated by students, on which they will reflect together. Draft session plan (in duos): before submitting the summative deliverable, students will get the chance to share and receive peer feedback on their session plan. The summative assignment (pass/fail) of this course is: Organize and facilitate a creative session for a real client (individual). This assignment will consist of 3 deliverables: Session Plan + substantiation of 2 activities (individual): to demonstrate that you can compile a session plan and know how to select relevant techniques. 2-pager Reflection (individual): demonstrating how well you reflect and integrate your knowledge and experience regarding Creative Facilitation, using the right principles and vocabulary. Client Report (individual): which is about properly and professionally communicating the session output to your client.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: cf-ide@tudelft.nl	

IDEM105	Exploring Design Intelligence	5
Course Coordinator	Dr. R.S.K. Chandrasegaran	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Language is the primary means of sensemaking and communication among designers, but is often overlooked. With recent developments in artificial intelligence (AI), language has become a dominant way in which designers co-perform with AI models such as Large Language Models (LLMs). Understanding language is thus crucial to develop future design practice. Exploring Design Intelligence (EDI) will provide you with a theoretical and practical foundation in the tools and methods with which to analyse design language and conversation. You will use this foundation as the basis for engaging with AI systems to model and generate design conversation, and produce AI design methods that can enhance design thinking capability. The course will help you develop technical and research competencies alongside deeper thinking about the process of design.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the features of design conversation. 2. Analyse design conversation using existing theories of designing and psycholinguistic tools and methods. 3. Create an AI design method or tool using text generation techniques	
Education Method	EDI uses a blended learning model where lectures, videos, and reading material will be used to provide students with relevant theories and approaches. Students will individually explore the material and reinforce their understanding of concepts through exercises and explorations in the classroom. Assignments will help students learn to use the tools and methods introduced in the course. Depending on the number of students, coaches will be assigned to student groups.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5318 Exploring Design Intelligence.	
Assessment	The study goals of this course will be assessed by a report (100%) There will be one main assignment in addition to in-class activity. This will be a project involving the collection and analysis of design conversation data followed by the creation and application of an AI model to a particular design scenario. Formative Assessment: Using the above learning activities as a basis, students will investigate and verify their data using the provided reading material, the provided manual/computational tools and techniques, and their own literature review. Their progress will be assessed by the coaches through individual or group meetings, and feedback given to the students. Summative Assessment: The main outcome from the assignment is a report of their investigation along with the outcome of the investigation (e.g. a method, or approach).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: edi-ide@tudelft.nl	

IDEM201	Advanced Visualisation for Communication	5
Course Coordinator	Ir. J.W. Hoftijzer	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Expected prior knowledge	Level of visualization knowledge and skills should at least reach 3rd year TU IDE Bachelor visualisation courses.	
Course Contents	This course focuses on the accurate and communicative visualisation (depiction) of information and processes, which concerns (design) narratives such as journeys, roadmaps, data visualisation and design proposals, etc., aligned to pedagogic goals. The course offers a general course-wide and integrating module, relevant to the different tracks of which each concerns a different focal area. Theory module The theory module will include a theoretical grounding with the help of literature, deepening lectures, instructions and optional excursions. The integration of theory connects theory of visualisation with the teaching and applying of visualisation knowledge and skills. The tracks concern graphic design, process/system visualization, customer and patient journeys, and technology and visualization (subject to change). Tracks Various tracks offer different ways and application areas of visualisation, aligned to the interests of students from all IDE Masters programs. The tracks will concern both analogue and digital formats and tools. The course will utilise software, tools and learning platforms in order to facilitate both contents and management (communication) of the course.	
Study Goals	Upon finishing this course, students are able to: 1. Explain relevant knowledge and theories of visual representation, communication and (sequential) stories. 2. Clearly and convincingly create a visual depiction of information. 3. Apply both general and track-specific theoretical principles and methods that enhance the clarity and convincingness of the visualizations. 4. Apply both general and track-specific visualisation techniques and tools efficiently and vigorously. 5. Evaluate a visualization and formulate in words and/or visuals suitable ways of improving/developing the design visualization.	
Education Method	Practical education with theoretic grounding In-class active learning; a mix of instruction, guidance, individual work, continuous feedback, peer feedback, each student: 1 block of 4 hours in-class (including break). In order to support learning we will, in addition to the 4 hours in-class teaching, offer facilitated self study(FSS) (= optional support or guidance). FSS is unscheduled consultation, offered on a voluntary basis, both helpful for students who need extra support and for those who seek extra challenges.	
Assessment	The study goals of this course will be assessed by: - Visual individual report (80%) - Individual exam (20%) 1. Visual report containing portfolio of all exercises, works & visualizations produced during the course and includes considerations, decisions, reflection 2. Exam (visual and written) that concerns theory, and a substantiated and well-formulated evaluation of visual work.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: avc-ide@tudelft.nl	
Remarks	For this course a maximum of 100 participants applies.	
Material List	Depending on chosen course track: drawing tablets and Adobe software are required.	

IDEM202	Advanced Visualisation for Design	5
Course Coordinator	Ir. J.W. Hoftijzer	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Expected prior knowledge	Level of visualization knowledge and skills should at least reach 3rd year TU IDE Bachelor visualisation courses.	
Course Contents	This course advances the students knowledge of, and skills in visualization principles, methodologies and techniques, aligned to pedagogic goals. This course offers a variety of tracks to choose from. Besides bringing knowledge and skills to a higher level, the course also broadens the students knowledge of visualization and offers methodologies and techniques that are highly relevant for exploring and expressing design ideas and/ or concepts. It covers a wide range of theories and topics concerning sketching and visualization. Theory The course addresses the visualisation components that concern the exploration of thoughts (sketching as thinking), communication with stakeholders, imagination, and the promotion/ facilitation of collaboration, aligned to pedagogic goals. The theoretic element will be offered through literature, deepening lectures, instructions and optional excursions. The integration of theory connects theory of visualisation with the teaching and applying of visualisation knowledge and skills. Tracks The individual tracks concern different design areas, and will concern mostly generative practical exercises, active learning while guided by experts. Each track will have a clear relation with the specific field of application and specific cases. Track topics concern e.g.: the human factor & interaction; visual creativity techniques; visualisation of mobility concepts; advanced design sketching (subject to change). Students will work with both analogue and digital tools and develop their understanding of generating visual depictions of complex objects, environments and people. Possible topics include: Theory: Spatiality, viewpoints and perspectives, rendering, light-dark (Chiaroscuro), Materials: properties and behaviour Colour theory Agency of sketching; methodology; Psychology of sketching; Psychology of perception; Skills: Development of motor skills, Advanced creativity- and ideation techniques; Visual exploration; The human factor; Human-product Interaction; Visualization for mobility (systems)	
Study Goals	Upon finishing this course, students are able to: 1. Explain relevant knowledge and theories of visualization as a thinking and development tool and language. 2. Create clear visual explorations of, and iterations on their concept sketch. 3. Apply both general and track-specific theoretical principles and methods that enhance the clarity and convincingness of the visualizations. 4. Apply both general and track-specific visualisation techniques and tools efficiently and vigorously. 5. Evaluate a visualization and formulate in words and/or visuals suitable ways of improving/developing the design visualization.	
Education Method	Practical education with theoretic grounding In-class active learning: a mix of instruction, guidance, individual work, continuous feedback, peer feedback, each student: 1 block of 4 hours in-class (including break). In order to support learning we will, in addition to the 4 hours in-class teaching, offer facilitated self study(FSS) (= optional support or guidance). FSS is unscheduled consultation, offered on a voluntary basis, both helpful for students who need extra support and for those who seek extra challenges.	
Assessment	The study goals of this course will be assessed by: - Visual individual report (80%) - Individual exam (20%) 1. Visual report containing portfolio of all drawings, sketches & visualizations produced during the course and includes considerations, decisions, reflection 2. Exam (visual and written) that concerns theory, and a substantiated and well-formulated	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: avd-ide@tudelft.nl	
Remarks	For this course a maximum of 100 participants applies.	
Material List	Depending on chosen course track: drawing tablets and Adobe software are required.	

IDEM203	Biomechanics	5
Course Coordinator	Dr. T. Huysmans	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Digital Human Models (DHM) are computer-based representations of humans that simulate interactions with virtual products or environments. Emerging in the 1960s, DHM has advanced significantly, especially with recent 3D scanning, AI, and computing innovations. It's a vibrant research field, vital for enhancing product function, fit, comfort, and safety in industries like automotive, aviation, defense, fashion, sports, and medicine. This course explores two key, intertwined DHM areas: 3D anthropometry and biomechanical modeling, particularly focusing on the musculoskeletal system (MSK). Initially, it provides an overview of human bones, muscles, and tendons, along with basic biomechanical principles. Students engage in workshops to learn about various DHM technologies, tools, and data. They will analyze dynamic human-product interactions, optimize them using DHMs, and set up physical studies to validate design choices. Example cases include designing sports equipment, bicycle seats, and respiratory masks. The course also covers tools, supports like chairs and beds, sports braces, wearables, and medical equipment.	
Study Goals	Upon finishing this course, students are able to: 1. Explain General Biomechanical Functions: Understand and articulate the roles of bones, muscles, ligaments, tendons, and joints in the human body, focusing on their biomechanical functions. 2. Apply Biomechanical Modeling Techniques: Use basic and advanced musculoskeletal modeling techniques for representing and analysing the biomechanics of human-product interactions. 3. Apply 3D anthropometric Modeling Techniques: Use state of the art methods, data, and tools for analysing fit in human-product interactions. 4. Propose Product Modifications to Optimise Human-Product Interactions: Examine human-product interactions, identify key influencing factors, and propose design modifications to enhance product function, fit, safety, and comfort. 5. Identify and Run Integrated Evaluation Approaches for Fit and Biomechanics: Develop and execute strategies for assessing fit and biomechanics in practical human-product interactions, selecting appropriate hardware, software, and criteria for effective evaluation.	
Education Method	Teaching through lectures, instructional videos, and hands-on workshops. Coaching of group assignments (4 students per group).	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5301 Biomechanics.	
Assessment	The study goals of this course will be assessed by: Formative assessment through coaching and peer feedback. Summative assessment through two group assignments: (1) OpenSIM tutorials (40%) and (2) Developing a Wearable Sizing System (50% report + 10% presentation).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: biom-ide@tudelft.nl	

IDEM204	Child and Play Perspectives	5
Course Coordinator	Ir. M.A. Gielen	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	How do children (0-12 years old) perceive the world? How do they make sense of it through their play? How can designers cater for these play needs? These questions are at the core of the course on Child and Play Perspectives. The course is a mix of weekly lectures, playful exercises, design assignments and reflective mini-essays. In a group, you will set up, execute and analyse two generative sessions with English-speaking children visiting our faculty. You will derive themes for play from these sessions. You will study core theory and practices of childrens play and perform a series of play exercises yourself. And you will use all the insights you gain to ideate on playthings (toys in the broadest sense of meaning). Through this course, you will gain understanding and competence in designing responsible play products for and with children. This is a playful course. You are invited to play along!	
Study Goals	Upon finishing this course, students are able to: 1. Explain and apply effective techniques for generative user research and co-design with children. 2. Derive and communicate insights on childrens perspectives from data gained in generative user research and co-design sessions. 3. Explain key concepts of theory on childrens play and relate them to play experiences. 4. Create design ideas and argue how they build upon theories of play and support specific play experiences in children.	
Education Method	Self-study + group practicals + reflections + discussion The course is built up around a series of learning blocks, each containing material for self-study, a related practical exercise to perform, a reflection assignment plus a concise implementation assignment (compact design exercise). Each learning block concludes with an in-class discussion. Next to this, two sessions with primary school children will be held and reflected upon. All reflections will undergo formative assessment during the course and can be improved before the final summative assessment.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5184 Co-design and Research with Children or ID5185 Design for Children's Play.	
Assessment	The study goals of this course will be assessed by: Formative assessment of individual reflections and group design exercises of each learning block are offered on a continuous basis, depending on students chosen moment of sharing preliminary work. The rubric will be utilized as the basis for feedback on formative deliverables. Summative assessment: One single summative assessment takes place at the end of the course, of two items: 1) final version of the series of individual reflections (mini-essays). 2) annotated design map (the series of co-creation findings and compact designs); this is group work. PASS/FAIL assessment is applied in this course.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: chip-ide@tudelft.nl	
Remarks	Due to collaboration with a primary school, a maximum of 32 participants can take this course.	

IDEM205	Cognitive Ergonomics for Complex Systems	5
Course Coordinator	Dr. R. van Egmond	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Cognitive ergonomics is traditionally concerned with mental processes, such as perception, memory, reasoning, and motor response, as they affect interactions among humans and other elements of a system. Relevant topics include mental workload, decision-making, performance, human-computer interaction, and skill acquisition. Nowadays, products and systems are becoming more and more intelligent by the use of artificial intelligence. This requires a completely new understanding of the interaction that humans have with these systems or the systems with humans. We postulate that intelligent systems and humans have different views and behaviors of the same world and that this presents a new challenge to design to find the proper balance. These topics will be related to and examined in the context of complex systems. The course consists of two parts: 1. A theoretical part; 2. An empirical part. Theory will be exemplified by current and past research of our faculty. The empirical part will be based on a real-life situation of an encounter with a complex system and an abstraction made of this situation. This abstraction should mimic an aspect of a real-life situation and be tested in an experimental setting. The final step is to derive the implications for the real-life situation.	
Study Goals	Upon finishing this course, students are able to: 1. Identify potential cognitive ergonomics problems in users interaction with a real-world complex system. 2. Abstract an experimental protocol from a real-world problem and test it in a laboratory setting. 3. Communicate the research outcomes in a scientific manner using appropriate visualization methods. 4. Develop design recommendations for supporting users' cognition in the interaction with the complex system.	
Education Method	The course combines lectures on core topics, group project work, and coaching sessions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5332 Cognitive Ergonomics for Complex Systems 1: Understanding Complex Information Systems or ID5333 Cognitive Ergonomics for Complex Systems 2: Application into Research and Design.	
Assessment	The study goals of this course will be assessed by a group assignment (100%) Formative: Three status update group presentations on problem identification, study setup, and data analysis. Summative: One assessment consisting of two parts: (1) Complete project presentation and (2) annotated presentation slides.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: cogergo-ide@tudelft.nl	

IDEM206	Contextmapping Skills	5
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Contextmapping is a design (research) method for including users and other stakeholders in a design process, being the expert of their experience. Underlying needs, motivations and dreams can be discovered by applying generative techniques in the research process. Over the past decade, the method, its principles and theories have been developed at TU Delft in close collaboration with Dutch and international researchers and practitioners. Contextmapping Skills goes in-depth in the fieldwork with coaching from expert practitioners at design agency Muzus and clients. Participation in Contextmapping Skills is open to all IDE MSc students who believe that end-users are important sources of information for leading new product development. DfI, SPD, IPD and PhD students.	
Study Goals	Upon finishing this course, students are able to: 1. Demonstrate and transfer empathic perspectives in both insights generation and insights delivery. 2. Design, conduct, and report an appropriate contextmapping study. 3. Analyse and organise data into relevant and meaningful insights. 4. Develop usable and impact-oriented deliverables. 5. Demonstrate open and constructive attitude and management.	
Education Method	In the course, teams of students perform a design research project for a client. The project follows the contextmapping process, at each step theory and instruction are provided in lectures, and literature is provided. At key moments presentations with feedback from peers, coaches, and experts take place. The course ends with presenting the findings to clients.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5663 Contextmapping Skills.	
Assessment	The study goals of this course will be assessed by: A final presentation and a report. Formative through weekly coaching and midterm assessment (report and presentation).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: cms-ide@tudelft.nl	

IDEM208	Design in Health: Theory & Practice	5
Course Coordinator	Dr.ir. S.N. Paus-Buzink	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Healthcare can be regarded as a complex system in which organizations, healthcare professionals, and patients interact with each other. In addition, the healthcare system is shaped by many rules and regulations. Developing healthcare products or services can pose specific challenges, for example due to the complexity of healthcare systems, the diversity of stakeholders involved or regulatory requirements. Within this course, you will learn about the different domains in healthcare, apply different research techniques and design methods, various perspectives on design for health and what your role as a Medisign engineer can be. Topics covered include (amongst others): - Ecosystem and roles of stakeholders within healthcare systems - Examples of role(s) of design(ers) within healthcare - Regulatory requirements for developing medical devices - Risk analysis & designing for safety - (Design) research in the clinical setting - Ethics in medical (design) research - Health technology assessment This course is highly recommended for all students following the Medisign specialisation.	
Study Goals	Upon finishing this course, students are able to: 1. Navigate the broad landscape of design for healthcare. 2. Interpret relevant procedures, rules and regulations on ensuring quality and safety of products and services for healthcare. 3. Integrate of research methods and ethical practices from the medical domain into design research. 4. Propose a redesign of a product, service or product-service system by applying methods and theories provided in the course.	
Education Method	The course includes weekly lectures and workshops, in which TUD experts and professionals from different areas within the field of medical design share their perspective. In the workshops, you will work in a small group (2-3 students) on assignment(s) connected to the lectures, which compile the final report to be submitted at the end of the course. Understanding of provided literature and lecture content and application of theories and methods will be assessed in an exam at the end of the course.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5721 Capita Selecta Medisign or ID5752 Rules and Regulations for Designing Medical Devices.	
Assessment	The study goals of this course will be assessed by: - Final group report of the assignments (40%) - Individual exam (60%)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: dhttp-ide@tudelft.nl	

IDEM210	eHealth Design for a Healthy Society	5
Course Coordinator	Dr. V.T. Visch	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	eHealth is often top-down technology-driven which has serious drawbacks on its uptake and negative effects on health disparity. This course takes a strict human-centered approach in the design of eHealth and has an overarching topic: 'How a human-centered approach of eHealth design can facilitate a healthy society' To this aim, the student will learn what eHealth is about, which specific design theories are relevant for it, and apply this knowledge in a practise specific case as well as in a general vision on the role of eHealth design for a healthy society. In this course students will be invited for the following health design journey in teams of 2: - Presentation of actual health innovation projects and design briefs by companies, external research groups, internal researchers. - Perform a critical analysis of interventions and create a preliminary future vision on the role of eHealth design for a healthy society. - Create design concept on design briefs - Present concept and vision to fellow students for peer feedback - Present vision and concept, including scenario, to stakeholders - Iterate vision and concept - Present report (including vision and concept) to tutors and stakeholders in final presentation. During the course interactive lectures will be provided on the following three perspectives of eHealth and the potential of design: - eHealth & healthy society (health philosophy, health-system and -value) - eHealth & health behavior (lifestyle, motivation) - eHealth & personalisation (groups (e.g. low SEP), profiling) The lectures will be given by faculty experts, guests, and PhDs/PDs. After each lecture, students will discuss with the expert the relevance of the topic to their health design project. Following these expert discussion, students will receive formative feedback on their project by the course coordinators. This course follows upon the elective Design in Health: Theory & Practice and is recommended for Medisign students.	
Study Goals	Upon finishing this course, students are able to: 1. Critically evaluate existing health interventions from a human-centered design perspective. 2. Apply the provided eHealth theory into the case-design by critically arguing for own design choices. 3. Create a convincing design concept based on eHealth theory for the given design brief by the client. 4. Create a vision on how eHealth design can contribute to a healthy society.	
Education Method	The students will work in pairs on a design brief that is formulated and introduced by a client. The course will have weekly meetings that involve a) presentations (by topic experts/ clients/ fellow students), b) discussions (with topic experts), c) formative feedback by individual group coaching by supervisors and/or clients.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5741 eHealth.	
Assessment	The study goals of this course will be assessed by a final report (100%) Each pair of students will be graded on their final report. The report will be written according to a template that involves the following parts: - Evaluation of provided health intervention. - Embedding created (re-)design in provided theoretical topics. - Description of created design. - Related vision on the role of design for Healthcare. Grading of the report is done by: - Embedding the created design in the respective lectured topics by the respective topics experts. - Evaluation and vision by coordinators.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: ehealth-ide@tudelft.nl	

IDEM212	Food and Eating Design	5
Course Coordinator	Dr.ir. H.N.J. Schifferstein	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Even though designers are specifically trained to create and build new products, their contribution to food innovation is still relatively small. This is a pity, because providing people with good quality, healthy, and appetizing food products that are produced in a sustainable way, and that take into account local differences in consumption contexts, habits and rituals is quite a challenge. Furthermore, the regional, seasonal, and perishable character of foods challenges designers to find solutions that connect consumers with agriculture, trading, and processing methods. The food & eating design course makes design students aware of the different fields of knowledge that are relevant for designs in which foods play a major part. The emphasis of this course is on bringing design students in contact with the knowledge areas that are specific for food and are not part of their regular design curriculum. The course addresses: agricultural production, food technology, sensory perception and aesthetics, health issues, packaging, marketing, food culture, food systems, food industry.	
Study Goals	Upon finishing this course, students are able to: 1. Describe the context in which food product development takes place 2. Analyze the context for production, acquisition, preparation, consumption, and disposal of a food product. 3. Apply and reflect on the knowledge from the course topics within a specific food category. 4. Develop innovations addressing multiple challenges for a specific food category.	
Education Method	Lectures, guest lectures, workshops, coaching sessions. Every week 1 or 2 topics are treated. Students work in pairs on their writing assignments every week. The design challenges start in week 3. The design challenges are individual assignments. The start of the course is focused more on the theory and the writing assignments, whereas towards the end of the course students will have more time to work on their design challenges and will receive feedback from peers or coaches on their challenges in coaching sessions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5295 Food and Eating Design.	
Assessment	The study goals of this course will be assessed by: - Group report (50%) - Design challenge (50%) In weekly meetings, students, working in pairs, will give and receive formative feedback from peers and coaches based on the same criteria used later to evaluate the summative assignment. The final deliverable consists of a group report, which is a collection of the improved written assignments. (50% of final grade) In addition, the students work on a design challenge individually. This will be presented through a poster, presented orally in class. The students submit their poster, together with a short motivation and explanation of their idea as part of this assignment. The design challenge makes up 50% of the final grade.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: fed-ide@tudelft.nl	

IDEM214	Lifestyle Research and Design	5
Responsible Instructor	M. Filippi	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This course focuses on sociocultural research and form-giving. It delves into the concepts of personal and group identity, and the notion of 'lifestyle.' Throughout the course, you will conduct field research to explore the preferences, values, and cultural nuances of a chosen (presumed) lifestyle group. Insights gained from this research will be integrated into the design process, shaping a product, service, intervention, or a combination of these. The aim is to create designs that are not only functional but also appropriate and meaningful within specific contexts and for a particular group of people. You will use form-giving as a process to seek design-user resonance and to craft compelling narratives, with the concept of form including, but not limited to its traditional vision-oriented scope.	
Study Goals	Upon finishing this course, students are able to: 1. Investigate lifestyle-specific preferences/values/attitudes/activities/behaviours through desk and field research. (ANALYSE) 2. Argue the validity of the chosen lifestyle group based on research findings. (EVALUATE) 3. Design a product, service, or intervention tailored to the specific preferences/values/attitudes/activities/behaviours of a chosen lifestyle group. (CREATE) 4. Communicate your research and design outcomes in an effective, coherent, and appropriate manner. (CREATE)	
Education Method	The course combines weekly: - Lectures on theoretical knowledge. - Hands-on exercises. - Coaching sessions (in project groups of 4-5 students).	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5293-18 Lifestyle Research/Design.	
Assessment	The study goals of this course will be assessed by: Groupwork, and consists of two assignments: - Assignment 1 Group project (research), (60%). - Assignment 2 Group project (design), (40%). Assessment of assignments will be based on two deliverables: a group process report and a group presentation/exhibition. Mention of individual contribution to groupwork and personal reflections should be included in the report. A peer formative feedback session will take place approximately halfway through the course. The assessment will be based on the course rubric.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: lrd-ide@tudelft.nl	

IDEM215	Lighting Design	5
Course Coordinator	Prof.dr. S.C. Pont	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	With advancements in technologies, new opportunities are available for application in (public) spaces to enhance (smart) environments, e.g. (O)LED lighting and screens, daylighting systems, emerging photonic (bio-)materials. These generate new possibilities to apply lighting in our public spaces for: - Ergonomic (e.g. visibility, perceptual and cognitive fluency, safety), - health (influencing biorhythm, recovery in hospitals, daylighting for wellbeing), - mobility (city master plans, lighting for transport facilities), - sustainability (lighting design for darkness, bio-design), - aesthetical (attractiveness, beauty, light art), and - commercial (branding, atmosphere creation, persuasive environments) purposes, - influencing our environment visually and non-visually and thereby our affects, cognitions, behaviour and (mental) health. In this interdisciplinary studio we work with academics and professionals covering various fields from the hard technologies to soft perceptions, namely lighting designers from BeersNielsen, Atelier LEK, Koerner Design, Philips, ERCO, over a dozen of Dutch lighting designers and professors from the faculty of Architecture. The lecturing lighting designers also provide opportunities for asking advice, and in the final stage the lighting design café is organised to provide such professional advice which currently attracts about a dozen practising lighting designers. The course offers training in light and lighting, both theory and practice of lighting design, via lectures and hands-on practicals, workshops and excursions, plus the popular lighting design café. Throughout you will be working on a lighting design assignment going through a complete design cycle researching, (re)framing, and formgiving the intangible matter called light. For whom, what, where, and when it is needed.	
Study Goals	Upon finishing this course, students are able to: 1. Assess light and its impact on humans. 2. Compose an integrated lighting plan. 3. Elaborate the lighting plan technically and describe, visualise, present (and pitch) the final lighting plan. 4. Explain how to extend the design process beyond humans.	
Education Method	The course is based on a strong theoretical grounding, and applies the theoretical learnings to the design or research practice through application exercises, and skill development exercises, while integrating design, scientific knowledge, and professional skills into a practical group assignment.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5115 Lighting Design.	
Assessment	The study goals of this course will be assessed by: - Final Design presentation (50%) - Final report (50%) The quality and content of the group assignment is assessed in a summative manner. Formative assessments In the first part of the course students will get four individual homework assignments on which they receive written feedback via Brightspace. The course organisers are permanent coaches during the course; mandatory group coaching meetings will be scheduled on specific course afternoons to provide oral feedback and voluntary coaching moments may happen on appointment. Halfway the concept for the group assignment will be pitched after which feedback by peers and teachers will be organized.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: ld-ide@tudelft.nl	
Remarks	For this course a maximum of 60 students applies.	

IDEM216	PerForm the Unseen	5
Course Coordinator	S.C.M. Brand-de Groot	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	In this course students will research unseen issues and render them tangible through (iterative) design processes in which formgiving the development, assessment, and determination of form takes centre stage. Designers can use their aesthetic sensibilities and communicative skills to support engagement with individual and collective concerns. Through their work, designers can spark and sustain public debate about these issues or revive controversies about long standing taboos. Matters of concern, moreover, of which many are intangible. Students will use an interdisciplinary set of skills to investigate how body language, individual perception and context influence the meaning of form in relation to design physical and/or digital artefacts. Students will investigate the communicative potential of form by means of embodied sketching and prototyping, experimental image and shape studies, observing and discussing. Students will develop their form studies into a design intervention that seeks to render tangible an unseen issue they chose to investigate. At the end of the course, they will evaluate the performance of their design intervention by means of a public presentation.	
Study Goals	Upon finishing this course, students are able to: 1. Analyse visual forms in their context using interdisciplinary skills like photographing, sketching and 3D modelling, artificial intelligence, discussing, philosophising. 2. Explore how they can use the human body as a ready yet rich source of information/inspiration for designing meaningful visions and designs. 3. Design with an intended meaning to make the intangible/invisible tangible, to bring about changes. 4. Communicate the vision of a design project by the conscious use of expression in form and performance. 5. Evaluate personal development, with reflections on insides and learnings.	
Education Method	Students discover the meaning of form by creating, observing, analysing, and discussing in groups. By using analogue (2D and 3D) and digital tools students will explore form by experience and doing. During the course we will be working with key skills like photographing, sketching, model making, artificial intelligence and bodily movements. In class we will do small individual and group assignments that we discuss in class. In the second half of the course, students will work in groups of 4 (or 5) students in a large design assignment including the performance at the end of the course.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5297 perFORM.	
Assessment	The study goals of this course will be assessed by: - Group assignment (40%) - Individual written exam (60%) During the course each week we will discuss with peers and teacher(s) to exchange experiences and outcomes. In an individual visual expression the student will perform the research, the design process, and the final outcomes/design of small and large assignments. The student must reflect and evaluate on the experiences, theory, and discussions in class. A final evaluation will be written. Next to the individual expression the students will be assessed on the group work: the design of a meaningful and coherent final form expression as an interdependent whole.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: perform-ide@tudelft.nl	

IDEM217	Strategic Automotive	5
Course Coordinator	E.D. van Grondelle	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This course aims to bridge between styling and strategy and make form giving strategic. The automotive industry is internally and externally undergoing unprecedented transformation. Designers must become strategic to embrace internal developments such as new propulsion technologies, new human product interaction, new ownership models and vastly increasing competition from new market entrants. Designers must also become systemic as car companies are transforming into ecosystems to adapt to the emerging holistic mobility paradigm which rebalances the design rationales for all mobility modalities. To accommodate this new design paradigm the automotive design process, which is traditionally rooted in tacit knowledge, requires a methodical framework. In this course we synthesise design methods and new styling management tools into a comprehensive set including tools and methods such as Form Strategy and Form Hierarchy, Buys, Porter and VR, Systemic Visualisations and the Metaverse. Note that while this elective's case studies and exercises are rooted in the automotive domain, its learnings are equally applicable in other (mobility) domains.	
Study Goals	Upon finishing this course, students are able to: 1. Students can analyse a brand's identity and image through a variety of secondary and primary sources. 2. Students can analyse product design (meaning of form) within the branding context, with respect to their underlying technology and business context. 3. Students can apply a variety of business models and methods to negotiate design with other disciplines and stakeholders and create a design strategy brief fit to automotive ecosystems. 4. Students can convincingly create a visual form language as the outcome of branding-by-design driven design brief.	
Education Method	Study load is spread evenly along the course, guided by case studies. This course consists of weekly 4-hour sessions, each of which combines - Short lectures (case studies), two or three per morning - Dialogue - Visualisation exercises - Project work - Impromptu presentations	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5249 Strategic Automotive.	
Assessment	The study goals of this course will be assessed by a final report (100%) Formative assessment by coaches and peers will be provided weekly in response to impromptu presentations. Summative assessment consists of one 3-member team assignment: a final presentation handout in the form of a report that includes processing of the feedback that was given upon the presentation. Students also write a personal evaluation, which may influence individual marks.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: strauf-ide@tudelft.nl	

IDEM218	Systemic Automotive Design	5
Course Coordinator	E.D. van Grondelle	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This elective belongs to our facultys strategic research theme Mobility. Automotive Design is no longer solely about personal vehicle design. Automotive Design is about applying its specific knowledge to mobility design in its full width, to rebalance individual wants and collective needs. We synthesise this tacit automotive design knowledge with the predominantly rational design paradigm of public mobility. In this elective, you will unravel the complexity of future mobility and society, to understand what meaningful mobility solutions are. By analysing and visualising THE CONTEMPORARY MOBILITY SYSTEM, you will identify what is relevant in todays context and identify your own position in the holistic system i.e., act as a systemic designer. You will develop a sense of urgency to explore what is appropriate within a future context. For that future context, you design an ADVANCED FUTURE CONCEPT VEHICLE, which is derived from your long-term future vision. How will mobility develop, and how should it be experienced? What is your futures personal, public, or shared vehicles reason for being? From there, you must design an INTERMEDIATE VEHICLE that bridges contemporary mobility to the advanced future concept.	
Study Goals	Upon finishing this course, students are able to: 1. Students can identify and explain their own position and power in the mobility domain and take a stand (systemic design). 2. The student can analyse a future mobility context in relation to a constructed future society, reorganise and correlate its factors into a holistic perspective, and construct a future vision. 3. The student can apply a context driven design process, balance between individual wants and collective needs, and translate its outcome into design manifestation. 4. The student can apply a variety of design tools, including AI, and convince stakeholders at a visual level as is expected in the automotive design domain.	
Education Method	This course is guided by a context driven design process which requires participants to keep pace. Weekly half day (4 hours) sessions include - Short lectures - Class discussion - Short progress presentations - Project work with coaching	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5242 Automotive Design and Mobility.	
Assessment	The study goals of this course will be assessed by: - Individual assessment 1 (50%) - Individual assessment 2 (50%) Study load is spread evenly along the course, guided by the context driven design process. The coherence throughout between analyses, storytelling, and manifestation, is fundamental. Weekly coaching project criteria will be informally and through dialogue addressed in class. Two individual summative assessments: an annotated presentation during week 5 addresses the storytelling aspect of the design project, and a presentation plus final visual in week 10, which assesses the manifestation i.e. the design.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: sad-ide@tudelft.nl	

IDEM219	Videography	5
Responsible Instructor	Dr.ing. M.C. Rozendaal	
Contact Hours / Week x/x/x/x	4	
Education Period	2 4	
Start Education	2 4	
Course Contents	Video, with its inherent narrative structure, visual and auditory richness, and ability to visualise future contexts, offers much potential for design and designers. In this course you will unlock some of this potential by learning how to use videography as a tool in your design process. You will be taught videography techniques and cinematic principles, and apply these in hands-on assignments. Following a general introduction, you choose one of three different tracks to deepen your skills: 1. Documentary Videography offers designers a way to capture the richness and detail of existing contexts of use, by showing real people performing real actions in real situations, while expressing genuine feelings, concerns and emotions. 2. Animation The potential of animation to combine imagery, music and sound in interesting new ways, offers designers a powerful medium to explain abstract ideas, convey a rich pallet of emotions, and simplify complex subject matter. 3. Fiction Videography offers designers the possibility to envision future design ideas through scenarios that communicate their form and use within particular contexts of use, and hereby addressing their potential impact in convincing and critical ways.	
Study Goals	Upon finishing this course, students are able to: 1. Apply videography skills in different stages of a video production. 2. Create quality media products for different design purposes. 3. Evaluate the value of videography in the design process.	
Education Method	The course will combine lectures and workshops, coach meetings, and make use of self-study. We start with a general (3-week) track in which students learn basic videography skills and fundamentals. Then in the following 6 weeks, students choose one of three different tracks to deepen their skills; (1) documentary (2) animation, (3) fiction. Students are consulted about their preferences a week before the course starts. Based on this, a balanced division across tracks will be made.	
Assessment	The study goals of this course will be assessed by: - Group portfolio (80%): Students build a video-portfolio based on group-work that will be assessed at the end of the course. - Written evaluation (20%): Students write an individual evaluation at the end of the course by reflecting on the value of videography in the design process.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: video-ide@tudelft.nl	

IDEM220	When Images Remain	5
Course Coordinator	Dr. M.W.A. Wijntjes	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	The course, titled When Images Remain, acknowledges the contemporary visual-centric world. It recognizes the significance of images as carriers of extensive, often invisible, information and explores the evolving role of visuals in human-machine knowledge transfer. The course unfolds the histories of image-making on a local and global scale, considering images as windows to the past and future, reflecting societal changes, artificial intelligence, and evolving values. This course delves into visual design through three distinct perspectives. First, it engages in theoretical analysis of existing image collections (see our archive), incorporating insights from art history, design, perception, and computer science. The critical examination considers both the creator's (sender) and the audience's (receiver) viewpoints, introducing the Vision and Depiction framework, which merges formal elements with contextual factors. Second, the course integrates empirical investigations. Students learn to conduct visual experiments, analyze data, and visually represent findings. The third perspective explores how computers interpret images, emphasizing the contribution of computational analysis to visual research and paying attention to ways in which visual research findings are shared and communicated.	
Study Goals	Upon finishing this course, students are able to: 1. Apply formal analysis based on the Vision and Depiction framework to gain in-depth experience into the expressive possibilities of visual interaction parameters. 2. Evaluate visual research using visual paradigms and computational approaches. 3. Reflect on the differences between human and computational interpretations of images, as well as the importance of developing an understanding of 'what makes an image' 4. Visually communicate the research results in a manner based on the Vision and Depiction framework.	
Education Method	Weekly lectures (2 hours), 3 workshops (2 hours) and 3 peer review moments.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4230 Visual Communication Design.	
Assessment	The study goals of this course will be assessed by an individual assignment (100%) All assignments are individual. Three formative assessments including the analysis and evaluation of images, and a reflection. The summative assignment will consist of the revised formative assessments, with the addition of a poster.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: wir-ide@tudelft.nl	

IDEM221	Making and Prototyping Skills	5
Course Coordinator	Dr.ir. T.J. Jakiewicz	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	In the prototyping elective, students will select from a relatively open range of small modules that cover different prototyping skills. Students start the course with a reflection on their professional identity as a designer, and individually identify 4 modules from the provided prototyping skill-tree that would support them in strengthening that identity. During the course, they go through the module-specific online tutorials (8h), and then spend time applying that knowledge to a project of choice, reflecting on and documenting their process (8h). Through most of the course students work autonomously. Support staff hold consultation hours where students can come and ask questions and discuss possibilities and strategies. SAs provide technical support. At the end of the course, students work in small groups to create a new or improved module tutorial. Negotiation: At the start of the course, students formulate their prototyping-skill ambitions and project-related prototyping needs. Course staff supports students strategically in selecting the right modules and collaborates with students on improving and developing new modules. Articulation: Students are responsible in the course for a combination of: - Redeveloping/improving an existing module to bring it up to date or add in additional ways that the skill can be developed. - Designing and testing a new module that teaches a new skill - based on talking with an expert, they formalise that knowledge, and make it communicable for future students.	
Study Goals	Upon finishing this course, students are able to: 1. Autonomously expand their own prototyping skillsets. 2. Select among various prototyping options based on skills, needs, and context. 3. Determine appropriate prototyping skills required in a project and critically evaluate the prototyping process. 4. Communicate how to use particular prototyping skills to others in an engaging and professional way and are able to contribute to practical knowledge generation exchange in a professional prototyping community.	
Education Method	Self-directed work based on online material. Optional weekly consultation moments in which students can obtain feedback and support.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5231 Experimental Formgiving of Visual Information.	
Assessment	The study goals of this course will be assessed by: Receiving a pass/fail grade based on submitting a) individual portfolio of filled in prototype card forms, and b) new tutorial or tutorial improvement submitted as a group. Submission of complete required deliverables, demonstrating autonomous completion of tutorials and new group contribution, guarantees a pass. Pass/fail assessment is performed by the core team at the end of the course with assistance of SAs. Students provide peer-feedback on each others tutorial contributions.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: maps-ide@tudelft.nl	

IDEM301	Animated Materials	5
Course Coordinator	Prof.dr.ir. K.M.B. Jansen	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This course is an introduction to the field of Animated Materials - materials that change during fabrication and/or use time, their relevance and application in the development of products, and the use of tools for the design of temporal materials and products made with them. Animated Materials facilitate transitions towards multi-situated, long-lasting artefacts that may enrich society and improve environmental impacts. A general introduction in a range of animated materials will be given, however, a selected animated material context will be explored in detail (for example: shape-memory materials, multimorphic textiles, e-textiles).	
Study Goals	Upon finishing this course, students are able to: 1. Discuss basic material components and examples of Animated Materials in terms of their context and impact. 2. Apply design and prototyping methods for Animated Materials. 3. Evaluate your design proposals and prototypes to determine relevant approaches. 4. Experiment with design and prototyping methods for Animated Materials. 5. Develop at least one Animated Material prototype/demonstrator that speculates a future application.	
Education Method	After a very focused exploration of fundamental materials and structures, the theories, tools and methods for (selected) animated materials will be conveyed through lectures and complementary workshops, in which students will have a chance to practise the given theory and techniques within teams of two or three. These lectures and workshops will be highly inspirational supported by recent example cases from the material science and design domain, and guest lectures where possible. Parallel to this the students will perform a short literature study on the performance and affordances of different classes of physically animated materials. After this, the students will conduct an experimental design research project to apply the theory they learned and create material concepts and applications. Herein, hands-on material experiments in the Lab (e.g. material tinkering, DIY material making, construction, modification), and studio discussions will be the main activities. While most of these activities will require self-study time, the experts will be available for individual meetings. Tools to support the development of materials that change will be delivered (low value physical prototyping, modelling, CLO3D for example) Week 1: Introduction to course, state of the art of Animated Materials (Lectures and discussion; instructions for benchmark study) Week 2: Animated Materials 1 Introduction to fundamental material properties of selected animated material (Lecture, examples, demonstrations and workshop) Week 3: Animated Materials 2 Deepening knowledge of Animated material (lectures and workshops) Week 4: Open Studio/Coaching Week 5: Midterm presentation (Results of Benchmark study and proposed direction for remainder of project) Week 6: Open Studio/Coaching Week 7: Open Studio/Coaching Week 8: Open Studio/Coaching Week 9: Open Studio/Coaching Week 10: Exhibition of prototypes and report submission	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5419 Multimorphic Textiles.	
Assessment	The study goals of this course will be assessed by: - Individual assessment (20%) - Group work (80%) There will be four main deliverables in the course: [1: 20%, individual] Benchmark report; [2: 20%, group] Prototype; [3: 20%, group] Video of prototype in future application scenario; [4: 40%, group] Project report, describing the design research process; The course will end with an exhibition in which students will present their prototypes and videos	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: am-ide@tudelft.nl	

IDEM303	Computational Fabrication and MetaMaterials	5
Course Coordinator	Dr.ir. E.L. Doubrovski	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	In this course we introduce a new way of thinking about and working with materials in digital design and 3D printing. Through computational fabrication, we can tailor material behaviour and experience. This is in contrast with common practice prescribed by traditional manufacturing, where materials are given or chosen. With computational fabrication, we not just consider the constraints of 3D printing, but actually leverage the specificities of the printing process to obtain structures with desired properties. To achieve this, a deep understanding of the inner workings of 3D printing is needed, as well as learning how the material responds to the 3D printing process. Blurring the boundary between 3D CAD and 3D printing, students will learn to use Grasshopper to define their designs as a set of machine instructions for the 3D printer. They will apply this to both Material Extrusion and Material Jetting techniques. Starting with simple structures, evaluating and testing their outcomes, and culminating in the application of their acquired knowledge to fabricate a prototype.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the differences between various multi-material printing processes 2. Apply a computational fabrication approach using Rhino Grasshopper 3. Analyse the effect of 3D printing parameters on the properties of a structure 4. Conceptualize and prototype a product in which the behaviour is achieved by leveraging the 3D printing process 5. Evaluate the 3D printed result, is the desired behaviour achieved?	
Education Method	Lectures, Workshops, Assignment	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5459 Computational Design for Digital Fabrication or ID5454 Digital Materials.	
Assessment	The study goals of this course will be assessed by: - Summative individual exam (50%). A computer-based exam, where students are asked to solve multiple tasks in Grasshopper. - Summative group assignment (50%). The groups are given a design task for which they devise a computational fabrication workflow, fabricate multiple prototypes for testing, and present the final design. The deliverables are a demo and written report.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: comfab-ide@tudelft.nl	

IDEM304	Design for Emerging Markets	5
Course Coordinator	Prof.dr.ir. J.C. Diehl	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This course is focussed on developing innovative product (service) systems solutions for emerging markets, in specific for low-resource settings. People in these settings are often confronted by a range of Sustainable Development Goal (SDG) challenges in their daily life such as a lack of access to basic healthcare (SDG3), food (SDG2) and education (SDG4), poverty (SDG1), and inequality (SDG10). While solving these challenges specific constraints of low-resource settings such as a lack of connectivity, infrastructure, low levels of literacy and limited (household) budgets have to be taken into account. In order to develop solutions for these SDG related challenges in low-resource settings, new design approaches and design mind-sets are needed like for example frugal innovation or design for the other four billion. These include deep listening to local consumers, new business models (like pay as you go), affordable but smart technologies, and new product / service infrastructures. In the first place this course will introduce the students to new research and design approaches for these emerging markets along a range of cases like smart health diagnostics, distributed renewable energy systems, and clean cooking. The student will be as well introduced to and prepared for the internationalisation of the industrial design profession, such as, entering international markets, international product policies, globalisation and working in international teams. Lastly, the students will reflect and debate about what the role of (international) industrial designers is in these contexts, as well as how should be taking into account aspects such as design ethics and decolonization.	
Study Goals	By the end of the course students will have an extended experience in analysing a low-resource context in an emerging country, able to apply context-specific design methods, and generate initial ideas for interventions. During the course the students will be provoked to reflect on their role and position as a designer in the context of low-resource settings. Upon finishing this course, students are able to: 1. Analyze and describe specific use-contexts and eco-systems in emerging markets. 2. Apply design methods and tools specifically for designing for emerging markets and low-resources settings (i.e Design for Well-Being, Frugal Innovation, context specific design). 3. Create innovative ideas for product (-service) solutions for low-resource settings in emerging markets. 4. Evaluate what the role and responsibilities are of designers active in emerging markets from an ethical perspective.	
Education Method	The course includes weekly lectures in which TU Delft experts and professionals from different backgrounds (NGO, business, non-profit consultancy, and academics) share their perspectives on Design for Emerging Markets. The lectures will be followed up by reflective discussions moderated by the course staff. The students will work in teams of maximum 5 students on a real life case with clients in the Global South. The students will be in direct contact with the client and weekly coached by the course staff. This will result into a final presentation and team report at the end of the course. The course will be wrapped with an individual reflective exam with open questions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5050-18 Design for Emerging Markets.	
Assessment	The study goals of this course will be assessed by: - Individual assignment (30%) - Group assignment (70%) Formative: Teamwork. Coach feedback using a rubric. Discussion after lectures. Feedback on involvement and input in the debate. Summative: (70%): a final team report and presentation of client based case that covers the analysis of the use-context, presents innovative solutions by applying context specific design methods. (30%): A reflective individual written exam on the role and position of designers in the context of low-resource settings.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: dfem-ide@tudelft.nl	

IDEM305	Designing Responsible AI	5
Course Coordinator	Dr. S. Colombo	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Designing Responsible AI equips students to design AI solutions guided by ethical principles and an understanding of societal impact. By combining AI technology with ethics and speculative design, students gain the skills to navigate AI development conscientiously, ensuring they contribute to the advancement of responsible AI applications. The course is structured around three core activities e 1. Analysis of AI-related ethical issues: Students dissect real-world AI use cases in sectors like healthcare or social networks, focusing on ethical implications, biases, and privacy concerns. This segment develops their ability to critically and ethically analyze AI technologies. 2. Design futuring for AI systems: This creative module pushes students to envision future AI technologies. They conceptualize near-future AI applications through speculative design/design fiction, balancing technological advancements with societal needs and ethical considerations. 3. Critiquing AI designs: Students present AI design ideas, engaging in discussions, assessments and critiques. This fosters a culture of constructive criticism, essential for refining responsible AI systems. The course emphasizes hands-on AI experience, ethical reasoning, collaborative projects, and expert insights. Students learn to design AI systems that are socially responsible and ethically sound.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the most significant ethical issues related to AI and several design principles and frameworks for addressing them. 2. Analyze real-world AI applications in context using responsible design frameworks and methods (e.g., value-sensitive design, guidance ethics). 3. Envision responsible near-future AI systems by applying design-futuring methods (e.g., Ethnographic Experiential Futures). 4. Critically evaluate the ethical (de)merits of responsible AI system design visions.	
Education Method	In this course, students will participate in interactive lectures, design activities, and debates to learn about the ethics and responsible design of Artificial Intelligence. During self-study hours, they will read assigned papers and reflect on them. - Lectures will focus on the ethics of the design of technology, AI in particular, as well as methods for responsible design of AI, such as value-sensitive design and design futuring methods. - Design activities will revolve around a group project where a speculative design of a responsible AI system is created. - During critiques, students will learn how to argue for the (de)merits of peers design proposals in a principled and structured manner. - The papers will cover topics relevant to the theories and methods discussed and applied during contact hours. Students will also occasionally perform preparatory exercises to generate materials that will be used in subsequent contact hours.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5417 Artificial Intelligence and Society.	
Assessment	The attainment of learning objectives will be assessed through one group and one individual assignment. 1. Group assignment (60% of final grade; artifact + presentation; Week 7-8): Students will deliver an artifact that embodies the design vision (animation, video, or experiential prototype demo). The artifact is accompanied by a presentation that describes the rationale (argument) for the design proposal based on ethical analysis of target design situation. 2. Individual assignment (40% of final grade; short report; Week 10): Brief written assessment report on a peer groups design proposal (~750 words). Includes a reasoned selection of one of several assessment frameworks and a discussion of the pros and cons of the design proposal and its accompanying rationale. Formative assessment will include: Case analysis (group): Applying a responsible design framework or method (e.g., VSD) to analyze a provided design case and extrapolate values. Self-assessing the quality of the analysis through a checklist. Critique exercise (group): Engaging in a critique exercise aided by a guide. Peers provide feedback.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: drai-ide@tudelft.nl	

IDEM307	Generative AI and Design	5
Course Coordinator	Prof. G.W. Kortuem	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	The course Generative AI and Design equips students to design and develop generative AI solutions and analyse their practical and ethical implications in a design context. Generative AI technologies such as Large Language Models (LLMs) and Latent Diffusion Models (LDMs), and generative AI tools such as ChatGPT and MidJourney, have the ability to generate content or data that is novel, creative, and often human-like. This course explores the impact of generative AI on design, and how we can use these new capabilities for creating new products and services, and new ways of designing.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the working principles and ethical dimensions of generative AI technologies. 2. Prototype feasible and desirable generative AI products. 3. Analyse the implications of generative AI for design.	
Education Method	The course will use a combination of - Lectures and demonstrations. - Student exercises with demonstrations and presentations. - Reviews, critiques and discussions. - Assigned readings.	
Assessment	The study goals of this course will be assessed by: - Individual exam (50%) - Group work report (50%) Formative Assessment: - Weekly quizzes: students will self-assess their understanding using provided model answers. - Weekly exercises. Feedback will be provided during in-class reviews and discussions. Summative Assessment: - Individual exam (50%) focusing on understanding of working principles and ethical dimensions of generative AI. - Group project (50%) where students will document a functional prototype of a generative AI product, and analyse implications of generative AI for design.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: gaid-ide@tudelft.nl	

IDEM308	Graduation Launchpad	5
Course Coordinator	Dr. J. Wu	
Contact Hours / Week x/x/x/x	4	
Education Period	2 4	
Start Education	2 4	
Course Contents	The Graduation Launchpad prepares students for a graduation project related to research of the IDE faculty (https://www.tudelft.nl/en/ide/research). The Graduation Launchpad acts as a platform where students can interact with researchers to identify topics for graduation projects and learn how to approach, plan and execute a successful graduation project. After a general introduction of how to conduct high-quality graduation work, students will join a topic-specific group of researchers, potential project supervisors and students to conduct preliminary research and prepare their individual graduation project. At the end of the Graduation Launchpad, each student will have gained a firm grounding in a research topic, defined a feasible graduation project, identified a supervisor team and be ready to start a graduation project. Note: Taken the aim and set-up of this course into account, it is logical to take this course in the third semester/the quarter before you start the MSc Graduation Project. Therefore, the course is offered twice a year, in quarter 2 and quarter 4.	
Study Goals	Upon finishing this course, students are able to: 1. Reflect on personal research and design competencies. 2. Demonstrate advanced understanding of a design research topic as foundation for a graduation project. 3. Develop a credible and feasible graduation proposal.	
Education Method	The course uses a mix of education methods including lectures, exercises, and researcher-guided self-study . In the first part, students will be taught how to conduct high-quality graduation work using lectures and classroom exercises. In the second part, students will conduct individual research guided by small teams of researchers and prospective graduation project supervisors. Students are able to choose who to work with based on a published list of potential graduation topics. Each Delft Design Lab and IDE research group / lab will be offered to inject topics into the graduation launchpad and guide students. Teams of researchers / supervisors may elect their own style of guidance. Guidance will be tailored by researchers and prospective graduation project supervisors with respect to graduation topics.	
Assessment	The study goals of this course will be assessed by an individual assessment (100%) (pass/fail) Individual assessment based on a final graduation project proposal.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: gral-ide@tudelft.nl	

IDEM309	Repair & Recycling!	5
Course Coordinator	Dr.ir. S.F.J. Flipsen	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Repair and Recycle! addresses product lifetime extension and material recovery. RR! enables students to apply dedicated design approaches to improve the reparability and recyclability of products. The way in which products are designed determines the result of the repair process (fault diagnosis, dis and re-assembly, part replacement, testing) and the recycling process (collection, liberation, sorting, reprocessing). For improving on repair and recycling in practice, also the social, economic, and legislative aspects will to be considered.	
Study Goals	Upon finishing this course, students are able to: 1. Analyse the reparability of a product considering technical and design aspects. 2. Analyse the reparability of a product considering social, economic, regulatory aspects. 3. Analyse the recyclability of a product (social, economic, regulatory, technical and design aspects). 4. Redesign a product to improve ease of repair and ease of recycling. 5. Identify and discuss tensions between reparability, recyclability, and reliability of a product.	
Education Method	The course will offer 6 introductory lectures, self-study material and 8 short workshops (lecture 1 h + workshops 3 hours contact time per week). Students apply the knowledge by assessing and improving the reparability and recyclability of a specific product.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5422 Repair!	
Assessment	Team report (100%) on the redesign of a specific product including - Product specific assessment maps (disassembly, hotspot, recycling). - Social, economic, and regulatory aspects. - Redesign for repair and recycling. - Demo object and validation. - Evaluation of tensions.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: rr-ide@tudelft.nl	
Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.	

IDEM401	BlueDot: Design Contest	3
Course Coordinator	Ir. J.G. Steenkamer	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Design a lowtech product on a given theme from concept to a detailed prototype to fit in the BlueDot collection.	
Study Goals	Upon finishing this course, students are able to: 1. Provide insight into their personal potential as an industrial designer. 2. Enhance their design skills and experience. 3. Develop a market-ready product. 4. Enhance their presentation skills. 5. Enhance their model building skills. 6. Experience in working with real deadlines. 7. Build on their personal portfolio and resume.	
Education Method	Weekly meetings with a mentor according to the timetable.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5548 BlueDot: Design Contest.	
Assessment	The study goals of this course will be assessed by an assignment (100%) The project is assessed by the mentor. For this he/she will receive a concise report with a personal reflection on the course objectives, necessary information regarding the design and photos of the model. (Models must be presented to the supervisor before submitting them to the contest.)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: blue-dot-ide@tudelft.nl	

IDEM402	BlueDot: Organisation	5
Course Coordinator	Ir. J.G. Steenkamer	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This project code is especially made for rewarding students who participated in the BlueDot Foundation as a member of the Contest Committee. Either they are paid through these ECTS points, OR they are given RAS maanden The Foundation BlueDot is a foundation for and by students with the objective to produce and market products designed by students. Every year the Foundation BlueDot has a new board and students may fulfill a role on that board. (Coursecode ID5552 / 6 ect) One of the activities that BlueDot develops each year is a Design Competition for students where the winners may - with the support of BlueDot - further develop and produce their design. The winning products are marketed by BlueDot. This contest is organised by the contest committee and students may fulfill a role on this organising committee. (Coursecode ID5552-17 / 3 ect) From BlueDots policy plan 2014/2015; Vision: The BlueDot Foundation strives to support all motivated students to develop their product design concepts into feasible and successful products. The Foundation thus creates a product portfolio, with which BlueDot can stand out as an innovative design label. The proceeds from sales are invested in new projects so that BlueDot can, financially independent, continue to support students. BlueDot Foundation aims to be a showcase of students' creativity and diversity of Delft University of Technology. Mission: To offer students a unique learning-experience by supporting them in the development of their product design concepts into a market-ready product.	
Study Goals	Study goals for students on the board of the Foundation BlueDot and/or the contest committee: - to gain experience with maintaining contact with producers and sellers - to gain experience with the principles of entrepreneurship - to gain experience with all facets of managing an organization - to learn how to fulfill marketing and promotional tasks - to learn how to fulfill selling tasks - to guide a production process - to learn about process management Furthermore, students learn about conflict resolution, communication, problem-solving, leadership and evaluation. The drawing up and maintenance of a schedule designed to accomplish aims within the set time. The learning-experience provided by being a member of the board of the BlueDot Foundation proves to be useful during studies and is much valued by future employers. For students who are active within BlueDot it is also possible to fill in their board- or committee year in a more personal way. Examples are board members who educate themselves in product photography, programming a website, marketing research or legal matters.	
Education Method	A personal learning process by being a member of the board or through the organisation of an activity, for instance organising the BlueDot Design Competition.	
Assessment	A personal written evaluation of the board- or committee tasks completed, with a reflection on the course objectives described above.	
Enrolment / Application	Interested in becoming a BlueDot board- or committee member? Make an appointment with the BlueDot chairperson. website: www.stichtingbluedot.nl email: info@stichtingbluedot.nl For TUD-IDE students only.	

IDEM403	Research (3 ECTS)	3
Responsible Instructor	P. Wang	
Education Period	Different, to be announced	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	Students participate, on an individual basis, in one of the ongoing research projects in the Faculty IDE. The student gets acquainted with setting up and executing a research project. Important activities are reading relevant literature, formulating research questions, data collection, data analysis and writing report or scientific article.	
Study Goals	After a research project, students are able to participate in ongoing research in the faculty and are able to execute a research project.	
Education Method	The research project is executed and supervised in a small group or on an individual basis. The project will be formulated within ongoing research in the Faculty IDE. A researcher working on this specific research will supervise the student/ small group.	
Assessment	The research project is completed with a report or paper. The supervisor evaluates this report or paper.	
Enrolment / Application	For TUD IDE-students only. Contact professor Sicco Santema for more information: S.C.Santema@tudelft.nl	

IDEM404	Research (5 ECTS)	5
Responsible Instructor	P. Wang	
Education Period	Different, to be announced	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	Students participate, on an individual basis, in one of the ongoing research projects in the Faculty IDE. The student gets acquainted with setting up and executing a research project. Important activities are reading relevant literature, formulating research questions, data collection, data analysis and writing report or scientific article.	
Study Goals	After a research project, students are able to participate in ongoing research in the faculty and are able to execute a research project.	
Education Method	The research project is executed and supervised in a small group or on an individual basis. The project will be formulated within ongoing research in the Faculty IDE. A researcher working on this specific research will supervise the student/ small group.	
Assessment	The research project is completed with a report or paper. The supervisor evaluates this report or paper. After the research project is done and approved by the supervisor, the grade, the course ID and the student ID are sent by email to s.c.santema@tudelft.nl, (Sicco Santema, the coordinator of the course). He will then put it in Osiris.	
Enrolment / Application	For TUD IDE-students only. Contact professor Sicco Santema for more information: S.C.Santema@tudelft.nl	

Year	2024/2025
Organization	Industrial Design Engineering
Education	Master Strategic Product Design

SPD Graduation Project (30 EC)

ID4390-16	Graduation Project (SPD)	30
Course Coordinator	R.A. Price	
Education Period	None (Self Study)	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	The graduation project gives the student the opportunity to show he is worthy of the Master of Science title and that he fulfils the requirements of the program. This implies that the student uses his knowledge and skills as an independent industrial design engineer to execute a complex project. The graduation project can be seen both as the masterpiece of the student, and as a (personal) learning experience. The scope is therefore not only on testing competences, but also on the development of knowledge, understanding and skills of the student during the project. The graduation project is considered to be a stepping stone to a future professional career. A high level of independence is therefore expected from the student in the planning and execution of the assignment and in the acquisition of knowledge, understanding and skills. A student may start the graduation project under the condition that he has passed all mandatory/first year courses of the Master's Program and the Project Brief is approved by the Board of Examiners.	
Study Goals	On completion of this course, the student will be able to: - Effectively collect, analyse, generate and evaluate knowledge required for the project - Justify his/her choices with respect to used methods and/or approaches used in the project. - Deliver a relevant (feasible, viable and desirable) project result - Effectively and thoroughly communicate to- and discuss with stakeholders involved in the project - Manage a design/research project independently within the given time	
Assessment	During the graduation project, there are three formal moment when the students' work is assessed; Mid-term, Green-light and final assessment. The assessment of the graduation projects is supported by means of a rubric. The deliverables for the graduation project consist of; - A report providing insight in the process and project results, - A showcase; either a prototype or model, a movie or a poster - A public presentation (at the graduation day) An evaluation of the project with the supervising team, consisting of a chairman and a mentor from IDE and, when applicable, an external mentor is scheduled directly following the public presentation.	
Remarks	A more elaborate description of the graduation project can be found in the document 'Graduation Manual'. (See; http://io.studenten.tudelft.nl/ - education - Graduation and Internship - Graduation (MSc)- Download - Graduation manual for students)	

IDEM3000	Graduation Project SPD	30
Course Coordinator	R.A. Price	
Education Period	None (Self Study)	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	This version of the graduation project is for students starting their master on or after September 2024. The graduation project gives the student the opportunity to show he is worthy of the Master of Science title and that he fulfils the requirements of the program. This implies that the student uses his knowledge and skills as an independent industrial design engineer to execute a complex project. The graduation project can be seen both as the masterpiece of the student, and as a (personal) learning experience. The scope is therefore not only on testing competences, but also on the development of knowledge, understanding and skills of the student during the project. The graduation project is considered to be a stepping stone to a future professional career. A high level of independence is therefore expected from the student in the planning and execution of the assignment and in the acquisition of knowledge, understanding and skills. A student may start the graduation project under the condition that he has passed all mandatory/first year courses of the Masters Program and the Project Brief is approved by the Board of Examiners.	
Study Goals	On completion of this course, the student will be able to: - Effectively collect, analyse, generate and evaluate knowledge required for the project - Justify his/her choices with respect to used methods and/or approaches used in the project. - Deliver a relevant (feasible, viable and desirable) project result - Effectively and thoroughly communicate to- and discuss with stakeholders involved in the project - Manage a design/research project independently within the given time	
Assessment	During the graduation project, there are three formal moment when the students work is assessed; Mid-term, Green-light and final assessment. The assessment of the graduation projects is supported by means of a rubric. The deliverables for the graduation project consist of; - A report providing insight in the process and project results, - A showcase; either a prototype or model, a movie or a poster - A public presentation (at the graduation day) An evaluation of the project with the supervising team, consisting of a chairman and a mentor from IDE and, when applicable, an external mentor is scheduled directly following the public presentation.	
Remarks	A more elaborate description of the graduation project can be found in the document Graduation Manual. (See; http://io.studenten.tudelft.nl/ - education - Graduation and Internship - Graduation (MSc)- Download - Graduation manual for students)	

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